

Gastroenterology

Sunday, November 30th, 2025

Dr. Alvi Islam



Overview

- **Disorders of the Liver**
 - Approach to elevated liver enzymes
 - Viral Hepatitis
 - Alcohol-associated Hepatitis
 - Metabolic-associated liver disease
 - Cirrhosis
 - Hemochromatosis
- **Disorders of the Intestines**
 - Celiac Disease
 - Inflammatory Bowel Disease
 - GI Bleeding
 - *H. pylori*
- **Disorders of the Esophagus**
 - Barrett's
 - Dysphagia
 - Achalasia, EoE
- **Disorders of the Pancreas**
 - Acute Pancreatitis
 - Chronic pancreatitis
 - Autoimmune Pancreatitis
- **BONUS SLIDES incl. Sample MCQs**

BONUS

Read on own

Guidelines

- CAG Guidelines: <https://www.cag-acg.org/publications/guideline-library>
- CASL Guidelines: <https://hepatology.ca/publications/guidelines/>
- AASLD Guidelines: <https://www.aasld.org/publications/practice-guidelines-0>
- ACG Guidelines: <https://gi.org/clinical-guidelines/clinical-guidelines-sortable-list/?sort=datedesc>
- AGA Guidelines: <https://gastro.org/clinical-guidance/>

Question 1: Liver — NEW 2026

A 27-year-old male with a history of heavy alcohol use presents with jaundice, ascites, and abdominal pain. He last drank alcohol 5 weeks ago. On exam, his temperature is 38.7 °C, blood pressure 95/60, HR 110. Labs: total bilirubin 112, AST 320, ALT 98, ALP 200, INR 2.0, creatinine 88. MELD score is 28, Maddrey Discriminant Function is 34. Ascitic fluid analysis shows PMN count 270 cells, culture pending. Abdo US show cirrhotic morphology but no biliary dilation. What is the next best step?

- A. Arrange for urgent ERCP
- B. Start IV Ceftriaxone + Albumin administration
- C. Order TJ liver biopsy for confirmation of histologic diagnosis
- D. Start Prednisolone + IV N-Acetylcysteine for severe alcohol-associated hepatitis

Pattern of Liver Enzyme Elevation

Hepatocellular (ALT, AST)

1000's: Toxins/Drugs (e.g. acetaminophen, cocaine – associated with ischemic injury)

Viral (Acute Hep A, B, D, or E)

Vascular: Shock, Budd Chiari

Acute stone w/ in 24 hours

Autoimmune hepatitis

Fulminant Wilson's (rare)

100's: Viral (Hep B, C, CMV, EBV), Congestive Hepatopathy, Alcohol Hepatitis AST > ALT

<100's: MASLD / MetALD, Celiac Disease, Hemochromatosis, Wilson's, A1AT, Hypothyroidism

Cholestatic (ALP, GGT)

Obtain US / MRCP

Extrahepatic (biliary duct dilatation):

- Painful → Stones
- Painless → Strictures (PSC), Benign obstruction (IgG4/AIP, AIDS cholangiopathy), malignant obstruction

Intrahepatic (no duct dilatation):

Drugs (antibiotics, TPN, estrogens, MTX), PBC, sepsis, cholestasis of pregnancy, infiltrative disease (e.g. sarcoidosis), malignancy/mets, congestive hepatopathy

Workup:

- Medication history!
- U/S, CT, MRCP, Endoscopic Ultrasound
- ERCP NOT for diagnosis, generally only for therapeutic intervention
- Consider liver biopsy

*Any of the above entities can cause mixed pattern of hepatocellular and cholestatic liver injury!

Hepatitis A

- Self-limited infection
- Fecal-oral transmission
- Diagnosis: Anti-HAV IgM
- Treatment is supportive, transplant for fulminant hepatitis
- You can immunize against it
 - Who?
 - Travelers to Hep A Endemic countries, chronic liver disease, MSM, PWID, recurrent plasma derived clotting factors, zoo-keepers or vets handling non-human primates
 - Post exposure prophylaxis: Household contacts of Hep A infected individuals, co-workers and clients of infected food handlers, contacts in childcare or JK/SK

<https://www.canada.ca/en/public-health/services/publications/healthy-living/canadian-immunization-guide-part-4-active-vaccines/page-6-hepatitis-a-vaccine.html>

Acute Hepatitis B

- **Diagnosis**

- Hepatitis B surface antigen (HBsAg) and core IgM are positive
- Test for co-infections: HCV, HDV, HIV, syphilis, gonorrhea, and chlamydia
 - HDV is now routine for all pts with HBsAg+ status; from most provinces it is a send-out test to national microbiology lab in Winnipeg

- **Natural history**

- <5% of adults progress to chronic HBV (re-check serology at 6 months for persistent HBsAg)
- <1% of adults develop liver failure

- **Treatment**

- Treatment is mainly supportive (>95% of patients recover completely)
- Consider antiviral therapy in short-term for fulminant hepatitis
- Ensure household and sexual contacts are immune and provide hepatitis B vaccine if they are not immune (anti-HBs < 10 mIU/mL)

Who should get Hep B Immunoglobulin? (CDC 2021) (in addition to vaccine)

Individuals who are unvaccinated or have unknown immune status with:

- Percutaneous (e.g. bite or needlestick) or mucosal exposure (sexual contact) to HBsAg-positive blood or body fluids
- Sex/needle sharing with HBsAg positive patient
- Victim of sexual assault by HBsAg positive assailant

Administer HBIG + vaccine preferably within 48h. Maximum interval unclear but likely 7-14 days

Who should get Hepatitis B vaccine?

- Universal immunization of infants recommended by CASL 2025 ideally at birth.
 - Vaccine schedules vary province to province (currently in Ontario – HBV vaccine in Grade 7)
- All infants born to HBsAg+ mothers at birth

Hepatitis B – Serology Patterns

Hep B Status	sAg	sAb	Core IgM	Core IgG	eAg	eAb	HBV DNA
Acute Infection	✓	-	✓	-	✓	-	✓
Window period	-	-	✓	+/-	+/-	+/-	+/-
Resolved Infection	-	✓	-	✓	-	✓	-
Chronic Infection	✓	-	-	✓	+/-	+/-	✓
Vaccinated	-	✓	-	-	-	-	-
Isolated Core Ab	-	-	-	✓	-	-	-

Chronic = Surface Antigen positive $\geq$ 6 months

Window period = Time between SAg disappearance and SAb appearance

Isolated Core Ab = can reflect false positive, remote infection with loss of sAb, or occult infection

Hepatitis B Miscellany (MCQ fodder)

EXTRAhepatic Manifestations of HBV

membranous
nephropathy >
MPGN, aplastic
anemia, cryo-
globulinemia,
Polyarteritis
nodosa.



INTERNAL
MEDICINE
REVIEW

Factors that ↑ risk of Cirrhosis

Host-related:

- ✓ Age >40
- ✓ Male sex
- ✓ Immunocompromised (e.g., HIV)
- ✓ Coinfection with HIV, HCV, or HDV
- ✓ Alcohol use
- ✓ Metabolic syndrome

Disease-related:

- ✓ High HBV DNA level (≥ 2000 IU/mL)
- ✓ Elevated ALT
- ✓ Prolonged time to **HBeAg** seroconversion
- ✓ **HBeAg-negative mutants**
- ✓ HBV Genotype C

***HCC can occur in Hep B in
the absence of cirrhosis!**

Factors that ↑ risk of HCC:

Host-related:

- ✓ Age >40
- ✓ Male sex
- ✓ Immunocompromised
- ✓ Coinfection with HIV, HCV, or HDV
- ✓ Alcohol use
- ✓ Metabolic syndrome
- Family history of HCC
- Birth in sub-Saharan Africa
- Aflatoxin exposure
- Smoking

Disease-related:

- ✓ High HBV DNA level
- ✓ Elevated ALT
- ✓ Prolonged time to **HBeAg** seroconversion
- ✓ **HBeAg-negative mutants**
- ✓ HBV Genotype C
- Persistent **HBsAg** positivity (vs. HBsAg clearance)
- Cirrhosis

Following Chronic Hep B – CASL/AMMI 2025 (NEW)

Who to screen for Hep B (HBsAg, HBsAb, HBcAb)

NEW *Universal, one-time screening for all adults ≥18

Specific High-Risk Groups (may need repeat/closer monitoring if negative):

1. Born in HBV endemic area
2. Household contacts of HBV carriers
3. HBV+ sexual partner
4. PWID
5. Incarcerated
6. CKD needing dialysis
7. Elevated liver enzymes
8. **All** pregnant women
9. Pts on immunosuppressive therapy

Baseline work-up for HBsAg+

- Physical exam (look for stigmata CLD & extrahepatic manifestations of HBV)
- ALT, CBC, Cr, HBV DNA, HBe serology, HIV, HCV, HDV
- Non-invasive fibrosis testing (Fibroscan or APRI [AST-to-platelet ratio index] in resource limited setting as per WHO 2024*)

Follow-up for HBsAg+

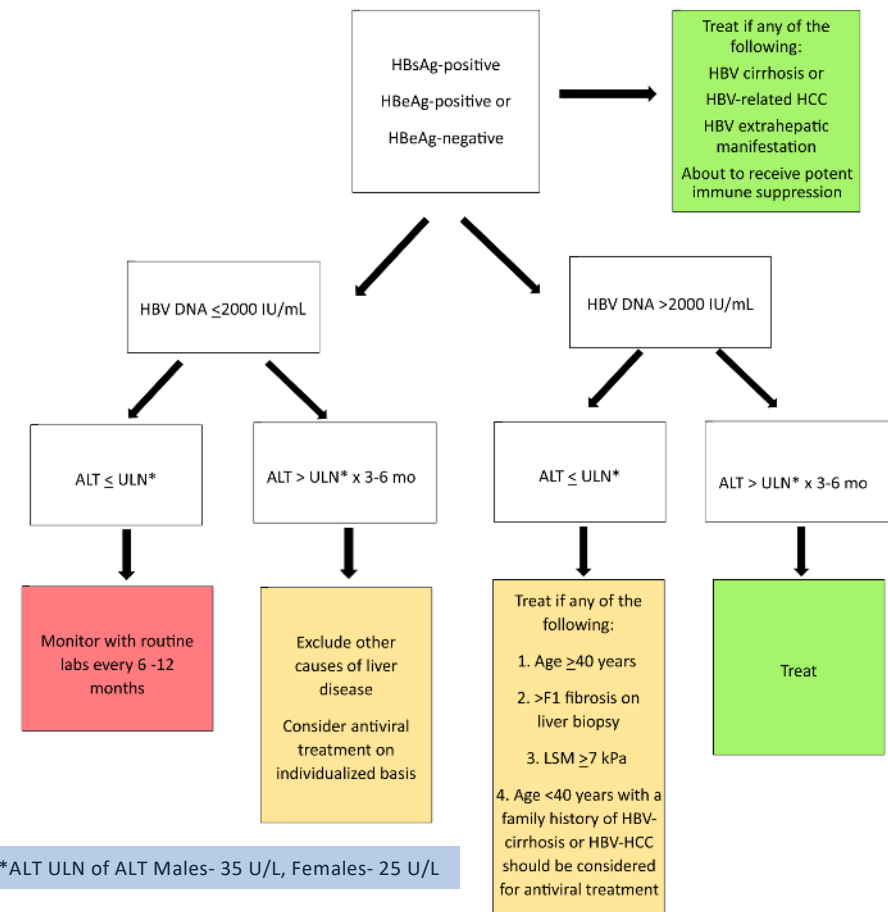
- ALT & HBV DNA q 6-12mos + Quantitative HBsAg q 1-2 years
- Repeat Fibroscan if persistent elevated ALT and HBV DNA
- **HCC Surveillance:** with Abdominal US + AFP q 6 months in following:
 - Cirrhosis
 - Person of African Ancestry ≥ 30y
 - Men aged ≥ 40
 - Women aged ≥ 50
 - Family History of HCC in first degree relatives – starting at age 40 or 10 years before affected family member (whichever sooner)
 - All HIV co-infected starting age 40
 - All HDV co-infected individuals starting at age 40, or earlier if advanced fibrosis >F3

Chronic Hepatitis B: Who to treat



CASL 2025 *UPDATED* TREATMENT INDICATIONS

1. Cirrhosis
2. HCC
3. *Pts requiring highly immunosuppressive therapy*
4. Extrahepatic manifestations of HBV
5. HBV DNA > 2000 IU/ml + $\uparrow$ ALT > ULN x 3-6 months
 - Regardless of HBeAg status (+ or -)
6. HBV DNA > 2000 IU/ml + any of:
 - Age ≥ 40
 - F2 fibrosis or higher on liver bx
 - LSM ≥ 7 kPa
 - Fam Hx of HBV-associated cirrhosis or HCC
7. Pregnancy/Pre-conception: HBV DNA > 200,000 IU/mL $\rightarrow$ consider immediate treatment, and no later than 2nd trimester (Goal to prevent MTCT)
 - Recommended tx = Tenofovir (TDF or TAF)
 - Tx up to 12 weeks post partum *if indication only to prevent MTCT
 - Infant should also get HBIG + HBV vaccine within 12h of birth and complete primary vaccination series before 6 months
 - Breastfeeding is not contraindicated



Treatment of Chronic HBV

Nucleotide Analogues (NA)

Tenofovir (TDF or TAF) or Entecavir (ETV) >> Lamivudine

- Tenofovir & ETV considered 1st line; very high barrier to resistance
 - Also recommended as 1st line Tx in WHO 2024
 - Generally, prefer TAF given less renal and bone A/E but \$\$
 - TAF or TDF can be used in pregnancy; Entecavir not recommended
- Potent viral suppression, but unlikely to seroconvert sAg
- Well-tolerated
- Indefinite therapy suggested for patients with Cirrhosis
- Usually life-long therapy in non-cirrhotic as well (especially in eAg negative disease)
 - In subset of patients tx can be stopped if HBsAg clearance (i.e. functional cure), after a 12 month consolidation period (but this is uncommon)
- Lamivudine is cheaper but high rate of viral resistance
- Adverse Effects:
 - Renal insufficiency (Fanconi syndrome / Type 2 RTA with TDF – check Cr and pH periodically)
 - Osteomalacia / osteoporosis
 - Lactic acidosis (rare)

Goals of treatment:

1. Prevent cirrhosis
2. Prevent HCC (which can develop without cirrhosis)
3. Achieve “Functional Cure” (i.e. defined as sustained loss of HBsAg or quantitative expression < 100 IU/ml AND undetectable HBV DNA)

Hepatitis C

Diagnosis	Anti-HCV	HCV RNA
Prior exposure & cleared (spontaneously resolved or cured by therapy)	✓	-
Chronic Hepatitis C	✓	✓

Natural History of Hepatitis C:

Acute infection (15-30% spontaneously resolve) → Chronic Hepatitis
→ Cirrhosis → Decompensation → HCC

Work-up at diagnosis:

- HCV RNA level, HCV Genotype Testing
- Liver enzymes, liver function tests, Cr
- Test for co-infections: HIV, HBV (HBsAg, anti-HBs, anti-HBc)
- Baseline Abdominal US in *everyone*
- Fibrosis Staging to exclude cirrhosis
 - TE/FibroScan, FibroTest, AST: Platelet ratio index (APRI), or biopsy as normal US does not exclude cirrhosis!

Who to screen for Hep C? (CASL 2018)

1. CASL= One-time population-based testing for anyone born between 1945-1975

- AASLD recommends universal screening for all patients ≥ 18
- SOGC recommends universal screening for pregnant patients

2. Risk-based Screening:

- History of past or current IVDU
- Received blood products before 1992 in Canada
- Born or lived in area w/ Hep C prevalence $>3\%$ (N. and sub-Saharan Africa; Middle East; Central, East, and South Asia; Eastern Europe; Australia)
- Born to Hep C + mother
- Current or previous incarceration
- HIV co-infection
- Sexual contact with HCV+ partner
- Elevated ALT
- Receiving hemodialysis

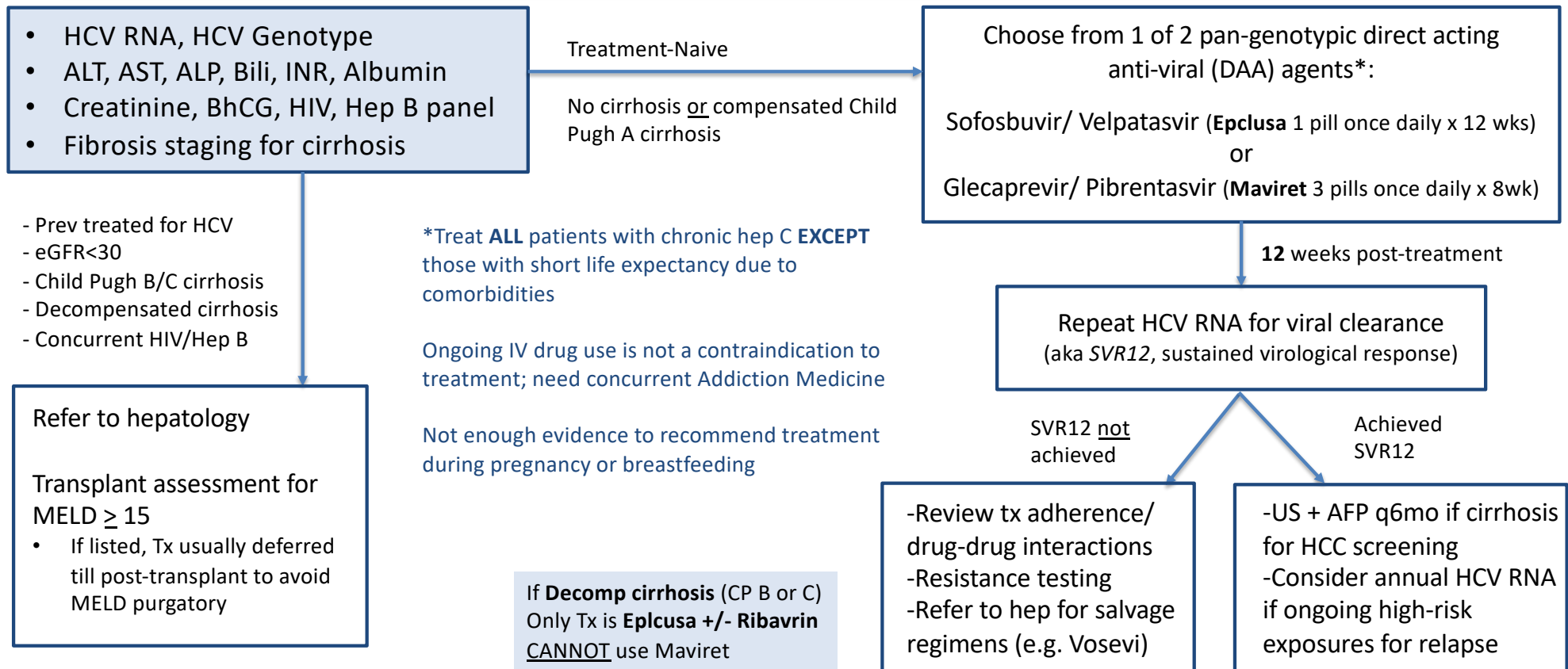
Annual re-testing of HCV RNA recommended for individuals with ongoing high-risk exposures e.g. HIV+, MSM, IVDU

MCQ fodder on Hepatitis C

- **Factors which increase risk of cirrhosis**
 - Older age, male sex, HIV/HBV co-infection, obesity/DM/fatty liver, alcohol abuse
- **Factors which increase risk of HCC**
 - Cirrhosis or co-existing liver disease which may accelerate fibrosis
 - Eradication of Hepatitis C infection does not fully eliminate HCC risk if patient has cirrhosis
- **Extrahepatic manifestations of hepatitis C:**
 - Autoimmune: Autoimmune thyroid disease, Sjogren's
 - Renal: MPGN > membranous glomerulopathy
 - Derm: Porphyria cutanea tarda, leukocytoclastic vasculitis
 - Heme: Lymphoma (NHL)*, autoimmune hemolytic anemia, ITP, **Cryoglobulinemia***
 - Other: Insulin resistance / Diabetes mellitus



Approach to HCV Treatment



Alcohol-Associated Hepatitis (AH)

- **Clinical Diagnosis for Probable AH:**

- Acute onset jaundice within 60 days of heavy alcohol use (>40g/day women & >60g/day men for ≥ 6 months); other causes of liver disease/jaundice excluded
- AST > 50, AST/ALT > 1.5-2, and both values < 400, Tbili > 51

Liver biopsy *rarely* needed.
Consider if 1. ALT, AST > 400,
2. Jaundice > 3 months, 3. ≥ 1
clinical criteria not met, 4.
Other dx suspected

- **Prognostication**

- Severe AH: Maddrey Discriminant Function (MDF) ≥ 32 or MELD > 20
- Infection associated with 3x $\uparrow$ risk of mortality (order septic screen: blood cx, urine cx, CXR, para to r/o SBP)

- **Treatment**

- EtOH cessation 
- Enteral nutrition (target = 35 kcal/kg/d and 1.2–1.5 g/kg/d of protein, consider thiamine & zinc supplement)
- Prednisolone 40mg daily x 28 days for severe AH or presence of encephalopathy
 - IV methylprednisolone 32mg daily is alternative for those unable to take PO meds
 - If 4 or 7-day Lille score < 0.45 $\rightarrow$ pred x 28d then taper; if $\geq 0.45 \rightarrow$ stop pred; consider early transplant (no longer need minimum period of abstinence; case-by-case) vs. palliative care
 - **C/I to steroids:** Uncontrolled infection, uncontrolled GI bleed, AKI
- IV N-Acetylcysteine (NAC)

NO role for the following
therapies:
-Pentoxifylline
-G-CSF
-Prophylactic antibiotics

Recommended by ACG 2024 as adjunct to steroids (improved 30d survival)

2024 ACG Guideline:
https://journals.lww.com/aig/fulltext/2024/01000/acg_clinical_guideline_alcohol_associated_liver.13.aspx
2019 AASLD Guidelines: <https://www.aasld.org/practice-guidelines/alcohol-associated-liver-disease>

Metabolic-Dysfunction Associated Steatotic Liver Disease

- NEW 2023 terminology for fatty liver disease: 
 - **MASLD** (NAFLD): Fatty liver with no hepatocellular injury
 - **MASH** (NASH): + hepatocellular injury
 - **Met-ALD**: MASLD + increased EtOH intake (20-30g/day females or 30-60g/day men)
- MASLD Diagnosis
 - 1) Evidence of steatosis (US, CT, MRI, histology, fibroscan)
 - 2) Rule out 2° causes (i.e. EtOH, drugs – steroids, methotrexate, amiodarone, estrogens)
- MASH can only be diagnosed definitively on Biopsy
- Associated with **T2DM** (most important risk factor), ↑lipids, HTN, obesity, metabolic syndrome, OSA, and endocrine disorders (PCOS, hypogonadism, hypothyroidism)
- MASLD is recognized as LEADING cause of chronic liver disease worldwide
- Most common cause of death in MASLD is cardiovascular disease
- Most cryptogenic cirrhosis likely “burned out” MASLD

Metabolic-Dysfunction Associated Steatotic Liver Disease

- Who to screen for clinically significant fibrosis ($\geq$ F2 fibrosis):
 - T2DM, obesity w/ metabolic complications, 1st degree relative with MASH cirrhosis, pts with alcohol use
 - Screen with **FIB-4 score** for everyone → if ≥ 1.3 , refer for fibroscan. If < 1.3 , follow with FIB-4 q1-3y depending on CV risk factors
- Treatments
 - **Weight loss** (diet + mod intensity exercise)
 - ≥ 3-5% ↓ to improve steatosis, 7-10% to improve fibrosis
 - **Identify & manage CV risk factors**
 - Statins are safe in MASLD/MASH, compensated cirrhosis
 - Complete EtOH abstinence if significant fibrosis ($\geq$ F2)
 - Coffee (≥ 2 cups/day associated with reduced risk of advanced fibrosis)
 - **Pharmacotherapy**
 - Resmetirom (Thyroid Hormone Receptor Beta agonist) –FDA approved in 2024 for bx-proven MASH with F2-F3 (Not approved in Canada yet)
 - Semaglutide (GLP1RA), tirzepatide (dual GIP/GLP1RA) off-label for MASH if pt has T2DM or obesity
 - Also studied: (1) pioglitazone for bx-proven MASH + T2DM, but risk of edema, CHF, (2) vitamin E 800 IU daily for Bx-proven MASH w/o T2DM, but may increase risk of adverse CV outcomes
 - **Bariatric surgery if BMI ≥ 35**
 - Increased post-op mortality w/ decompensated cirrhosis or clinically significant portal hypertension

$$\text{FIB-4} = \frac{\text{Age (years)} \times \text{AST (U/L)}}{\text{Platelet Count (10}^9\text{/L)} \times \sqrt{\text{ALT (U/L)}}}$$

Approach to Cirrhosis: Causes

- **Determine etiology**

- (1) Viral (2) Steatotic liver disease [EtOH, MASH], (3) Autoimmune/cholestatic [AIH, PBC, PSC], (4) Genetic [Wilson, HH, A1AT], (5) Chronic biliary disease, (6) Vascular [R CHF, Budd Chiari]
- Labs, abdominal imaging (+ doppler), and consider biopsy if indeterminate cause

- **Assess severity**

- **Child-Pugh score** – predicts perioperative mortality for open abdominal surgery (based on old data, likely overestimates risk):
 - CP A (score 5-6) – 10%
 - CP B (score 7-9) – 30%
 - CP C (score 10-15) – 80%
- **MELD ≥ 15** – refer for liver transplant assessment
 - 6 months of EtOH abstinence no longer a rule in Ontario (case by case)
- For perioperative liver-related mortality, can use the **Mayo clinic surgical risk score** or the **VOCAL-Penn score** (both online, more refined + current than the Child-Pugh score; but know CP for exam)

Child Pugh Score

***Don't memorize, but helpful to have a sense of 'A' vs 'C'!**

Ascites

- Absent (1), slight (2), moderate (3)

Bilirubin

- <34.2 (1), $34.2 - 51.3$ (2), > 51.3 (3)

Coagulopathy (INR)

- < 1.7 (1), $1.7 - 2.2$ (2), > 2.2 (3)

Albumin

- >35 (1), $28 - 35$ (2), < 28 (3)

Encephalopathy

- None (1), grade 1-2 (2), grade 3-4 (3)

Approach to Cirrhosis: Complications + Counseling

- **Assess for complications = “Decompensated” cirrhosis:**

Varices	EGD at diagnosis (if Fibroscan not available) or time of decompensation, then interval based on findings
Ascites/SBP	Paracentesis at diagnosis of new ascites, any inpatient presentation, and if concern for SBP
Hepatic encephalopathy	History / exam
Hepatorenal syndrome, hepatopulmonary syndrome / porto-pulmonary hypertension	History / exam / labs
HCC	Abdominal US / AFP at diagnosis and q6 months thereafter

Approach to Cirrhosis: Complications + Counseling

- **Counseling:**



- Complete abstinence from alcohol, adequate nutrition, weight loss for MASLD
- Limit acetaminophen to $\leq 2\text{g/day}$
- Avoid **sedatives, NSAIDs, ACEI/ARBs**
- ACG 2025 Nutritional Recommendations:



1. **Coffee** consumption to reduce fibrosis progression,
 2. **Branch-chain amino acids** (BCAAs) in addition to standard therapies for Hepatic Encephalopathy (HE)
 3. late evening **high protein snacks** b/w 7-10pm to reduce risk of ascites and HE
- Screen & refer to multidisciplinary team if frailty, sarcopenia or malnutrition (AASLD 2021)
 - Vaccinations
 - HAV, HBV, COVID 19, TdAP, Pneumococcal, flu vaccinations
 - +zoster if age >50 , MMR & Varicella where applicable, RSV if over age 60
 - Contraception / pregnancy planning in reproductive-aged patients
 - Estrogen-containing contraception and menopause hormonal therapy should be **avoided** in decompensated cirrhosis, Budd-Chiari, or hepatocellular adenoma
 - Refer to palliative care team if decompensated cirrhosis at any point in journey
 - → improves symptoms, quality of life, caregiver stress

2023 AASLD Guidelines: <https://www.aasld.org/practice-guidelines/clinical-assessment-and-management-non-alcoholic-fatty-liver-disease>

2021 AASLD Reproductive Health & Liver Disease Guideline

AASLD 2022 Guidance statement on Palliative Care

2025 ACG Malnutrition & Nutritional recommendations: https://journals.lww.com/ajg/fulltext/2025/05000/acg_clinical_guideline_malnutrition_and.13.aspx

Evaluating Risk of Decompensation in Cirrhosis

- **Decompensated cirrhosis – high risk, implies clinically significant portal HTN**
 - Perform EGD at time of decompensation
 - If high-risk varices identified, treat with non-selective beta-blocker or endoscopic variceal ligation (NSBB > EVL)
- **Compensated Cirrhosis – evaluate for Clinically Significant Portal Hypertension (CSPH)**
 - **CSPH** = Hepatic venous pressure gradient (HVPG) ≥ 10 mmHg = $\uparrow$ risk of hepatic decompensation
 - **Measuring CSPH:**
 - Invasively – during transjugular liver biopsy
 - Non-invasively – (i) *transient elastography* (TE - eg) Fibroscan - gives (“liver stiffness measurement”) + (ii) platelet count
 - LSM ≥ 25 kpa **or**
 - LSM ≥ 20 kpa + Plt < 150 **or**
 - LSM 15-19.9 kpa + Plt < 110
 - If Fibroscan is not available, other surrogate markers of CSPH include:
 1. Presence of portosystemic collaterals or hepatofugal flow on imaging
 2. Evidence of varices on imaging
 3. Clinical decompensation
- 1st line TREATMENT = Non-Selective Beta Blocker (NSBB $\downarrow$ risk of future decompensation)

Evaluating Risk of Decompensation in Compensated Cirrhosis

Moving away from EGD for everyone

Compensated Cirrhosis

No CSPH

- LSM < 20 + plt ≥ 150
- No varices on EGD
- No portosystemic collaterals on imaging

- Patient at low risk for varices; no need for NSBB but need to monitor periodically for development of CSPH
- If TE is available:
 - Monitor LSM + plt q1y for CSPH
 - If patient develops CSPH, start NSBB
- If TE not available:
 - Monitor with EGD q2-3years
 - If varices identified, patient has CSPH → start NSBB

***High-risk varices:**
mod-large size, red wale marks, or Child-Pugh C

Clinically Significant Portal HTN (CSPH)

LSM ≥ 25kpa or LSM ≥ 20kpa + Plt < 150 or LSM 15-19.9 & plt < 110
Varices of any size on EGD
Portosystemic collaterals on imaging

NSBB (carvedilol preferred)

Adherent to NSBB

No further EGD

Intolerant to or C/I to NSBB

EGD surveillance to detect varices

- q2y if ongoing liver injury
- q3y if no ongoing injury

*High risk varices identified

Low-med risk varices identified

EVL
q2-4wks until varices obliterated

EGD q1-2y depending on if ongoing liver injury

Once eradicated
EGD q6-12mo

Management of Varices in Decompensated Cirrhosis

- **Primary prophylaxis** = no prior acute variceal hemorrhage (AVH) ★
 - Patients receive: EVL OR NSBB
 - AASLD 2023 preferentially recommends NSBB over EVL, reserving EVL for those who cannot tolerate NSBB or have contraindications
- **Secondary prophylaxis** = after AVH
 - Patients receive: EVL (within 12h of bleed) AND NSBB*
 - *Exception: In rare cases, patients may undergo pre-emptive TIPS as 1st line therapy for AVH. If so, no need for secondary prophylaxis with NSBB
 - Inpatient setting: IV octreotide x 2-5 days → transition to NSBB once octreotide DC'ed, ideally within 7d of admission

EVL: Endoscopic Variceal Ligation
(aka variceal banding)

- Requires serial EGD w/ banding q2-4 weeks to eradicate varices and then ongoing surveillance q6-12months

NSBB: Non-Selective Beta Blocker

- Carvedilol preferred over nadolol and propranolol (AASLD 2023)
- Carvedilol starting dose = 3.125mg BID and titrate up to maintain sBP ≥ 90 (no HR target)
- Nadolol & Propranolol: titrate to HR 55-60 and maintain SBP ≥ 90
- Contraindications to NSBB (AASLD 2023)
 - 2nd/3rd degree AV Block (no PPM)
 - Sick sinus syndrome
 - Bradycardia (HR < 50)
 - Asthma (severe uncontrolled)

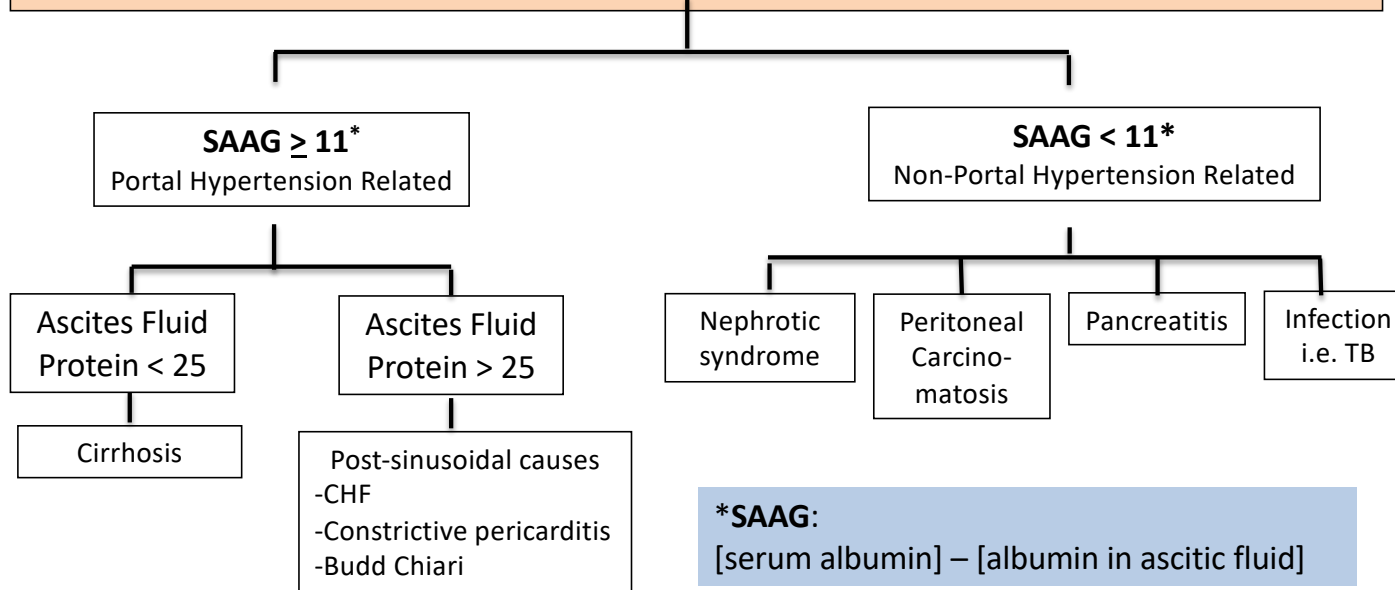
Ascites

EVALUATION OF NEW ONSET ASCITES

History: risk factors for liver disease, CHF, malignancy, pancreatitis, hypercoagulable disorders; travel hx/risk factors for TB

Exam: stigmata of liver disease, heart failure or constrictive pericarditis (JVD, pulmonary congestion, pericardial rub), signs of malignancy or infection (lymphadenopathy), malnutrition (sarcopenia), signs of thyroid disease

Investigations: Liver enzymes + function, lytes & Cr, urine protein, TSH, abdo US w/ doppler, echocardiogram



***SAAG:**
[serum albumin] – [albumin in ascitic fluid]

Initial Management of Ascites Secondary to Cirrhosis

- Treat underlying liver disease
- Avoid NSAIDs, ACEi, ARBs
- Salt restriction <2g or 88 mmol/d
 - Water restriction not necessary unless [Na] < 125; then fluid restrict ≤ 1L daily
- Diuretics
 - Starting dose = Spironolactone 100mg & Lasix 40mg, titrate as tolerated w/ interval of at least 72h
 - Max dose = spirono 400mg, Lasix 160mg
 - Monitor for hypotension, AKI, orthostasis, electrolyte imbalance

Refractory Ascites

- Assess adherence to diuretics & Na+ restriction (see next slide)
- Large volume paracentesis
 - Give albumin 6-8 g/L of fluid removed for taps > 5L
- TIPS if no contraindications (encephalopathy, HCC, pulm. HTN, R. HF)
- Liver transplant

Ascites management

- Patients with ascites should be Na restricted (<2g/day = 88mmol/day)
- 10mmol/day of non-urinary Na loss; therefore:

1) For patients not on diuretics, do 24-hour urine Na collection

- 24-hr urine Na <78 mmol → COMPLIANT w/ diet
- 24-hr urine Na >78 mmol → NON-COMPLIANT

2) Start diuretics if ongoing ascites or unable to comply

3) For patients on diuretics, do 24-hour urine Na collection

Can use spot urine Na/K ratio ≥ 1 as surrogate for 24h urine Na > 78 mmol, but gold standard (eg on exam) remains 24h urine Na

24h urine Na result	Interpretation	Treatment
<78 mmol	Diuretic resistant at current doses	↑diuretic dose until limited by AKI, electrolyte abnormalities, BP etc.
>78 mmol AND patient <i>not</i> losing weight	NON-COMPLIANT with sodium restriction	Reinforce sodium restriction
>78 mmol AND patient <i>has</i> weight loss	Adherent to Na restriction and diuretic sensitive	Stay the Course!



Spontaneous Bacterial Peritonitis

Dx: Neutrophils in ascitic fluid ≥ 250 cells/ml **OR** culture-positive ascitic fluid. Rule out secondary causes of peritonitis (i.e. bowel perf, recent surgery, intra-abdominal abscess/infections)

- **Treatment of confirmed SBP**

- Ceftriaxone (or Fluoroquinolone if CTX allergic) x 5 days
- HRS prophylaxis → **Day 1:** Albumin 1.5g/kg; **Day 3:** Albumin 1 g/kg
 - In practice everyone gets albumin, but guidelines suggest greatest benefit if Cr > 88, BUN > 10.7, or bili > 68
 - Post SBP: lifelong prophylaxis with Norfloxacin, Septra DS, or Cipro
- NSBB should be temporarily held in patients with SBP + hypotension or AKI

This is **the only** indication for albumin in clinical medicine that has a **mortality benefit!**



- **Indications for SBP prophylaxis**

- 1) Patients who have previously had SBP → indefinite SBP prophylaxis
- 2) Patients with cirrhosis who present with any upper or lower GI bleeding (*don't need to have ascites, doesn't need to be a variceal bleed*) → SBP prophylaxis x **5 days**
 - AASLD 2023: "Antimicrobial therapy can be discontinued once bleeding is controlled and in absence of an active infection"
- 3) Cirrhotic with ascitic fluid protein is <15 g/L and at least one of:
 - Impaired renal function (Cr ≥ 106 , BUN ≥ 8.9 , Na ≤ 130)
 - Impaired liver function (Child-Pugh ≥ 9 and Bili ≥ 51 umol/L)



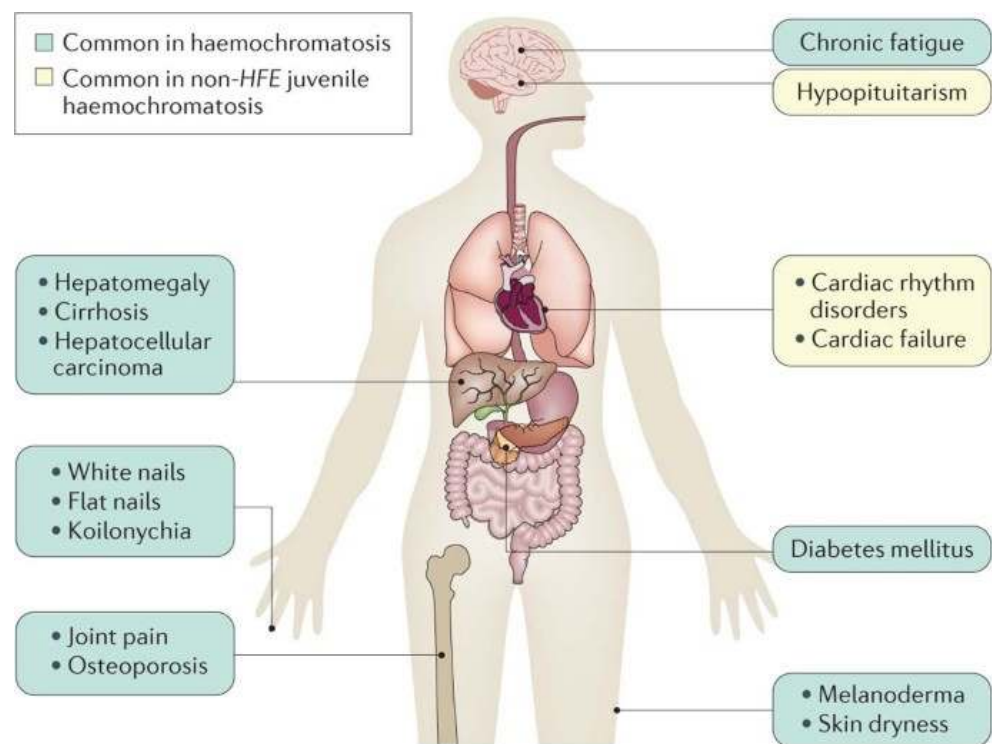
Hereditary Hemochromatosis (HH)

ACG Clinical Guideline: Hereditary Hemochromatosis (2019)

- **Autosomal recessive** disorder causing ↑ **intestinal absorption of iron** → **iron overload**
 - More common in Northern European descent
 - Most HH is due to **mutation of HFE gene at C282Y, H63D, or S65C**
 - Type 1A hemochromatosis: C282Y homozygous = confirms diagnosis of HH
 - Type 1B hemochromatosis: C282Y/H63D heterozygous = less likely but possible to have iron overload
 - Genotypes that do not lead to clinically significant iron overload: C282Y/ wildtype, H63D/ wildtype, H63D homozygote, S65C mutation (type 1C hemochromatosis)
- **S65C mutation** can cause slightly elevated ferritin/iron sat. Not clinically important that is why you won't see this on reports in Canada. Even if C282Y/S65C heterozygote, iron overload-related liver disease hasn't been reported.
 - Non-HFE-related HH (type 2- 4 HH) is very rare; not on exam in previous years

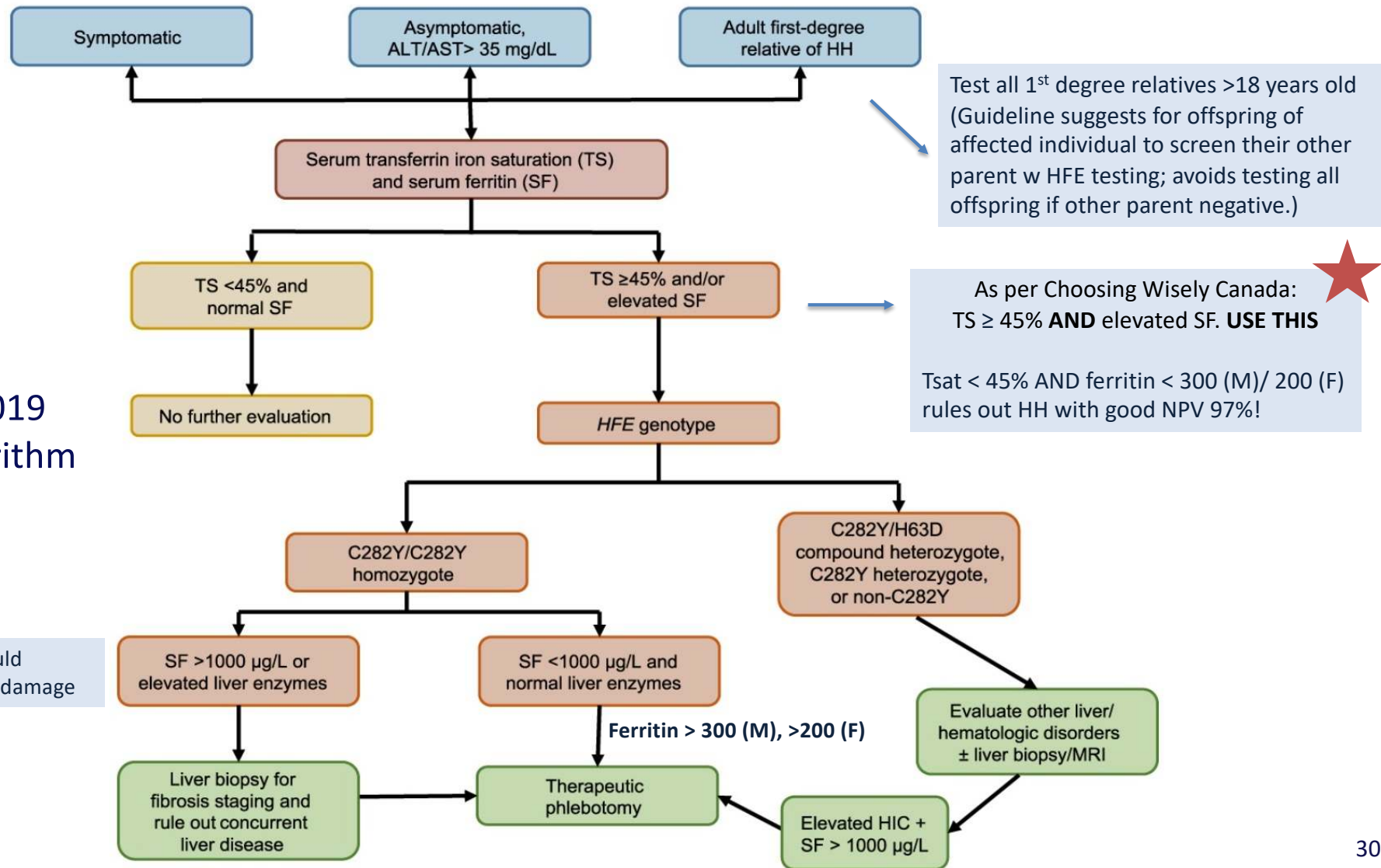
HH Clinical Manifestations

- Iron deposition in end-organs:
 - **Liver:** Cirrhosis, elevated liver enzymes
 - **Skin:** 'Bronze' hyperpigmentation
 - **Endocrine:** Hypopituitarism (impotence, amenorrhea), diabetes (islet cell destruction).
 - **MSK:** Arthropathy (2nd/3rd MCP hooked osteophytes)
- Fatigue
- Increased risk of infection: *Listeria monocytogenes*, *Yersinia enterocolitica*, *E. Coli*, *Vibrio vulnificus*.
- **DDx Iron Overload (secondary iron overload)**
 - Dyserythropoeisis (sickle cell, thalassemia)
 - Chronic transfusions
 - Other chronic liver diseases:
 - MASLD *RC favorite
 - Alcohol-related liver disease
 - Hepatitis C



Nature Reviews | Disease Primers

ACG 2019 HH Algorithm



HH Treatment : De-ironing



- **Who + when to treat:**

- C282Y homozygotes if ferritin > 300 (men) or > 200 (women) AND Tsat > 45%
- C282Y/H63D heterozygotes/ other → Evaluate for other causes of iron overload/ liver dz + consider MRI/ Bx to estimate hepatic iron concentration (HIC) → Treat other causes 1st, then, if high HIC → treat

- **Treatment:**

- 1st-line = phlebotomy target ferritin < 50 for induction → ferritin 50– 100 during maintenance
- 2nd- line (when patient refractory to phlebotomy e.g. anemia, high output CHF) = chelation (i.e. desferoxamine, deferiprone, deferasirox)
 - Risks of retinal/auditory toxicity (desferoxamine), agranulocytosis (deferiprone), liver/renal toxicity (deferasirox).
- No need to limit red meat/dietary iron if undergoing phlebotomy
- Avoid consuming vitamin C supplements (↑iron absorption)
- Avoid uncooked seafood (Listeria, Yersinia, and Vibrio ♥ Fe)
- Liver transplantation for decompensated cirrhosis/focal HCC

- **HCC screening:** Only if cirrhosis. ACG recommends *against* screening if fibrosis ≤ stage 3.

- **Other Heterozygotes:** Monitor iron indices annually + assess for organ damage, no treatment needed

Question 1: Liver

A 27-year-old male with a history of heavy alcohol use presents with jaundice, ascites, and abdominal pain. He last drank alcohol 5 weeks ago. On exam, his temperature is 38.7 °C, blood pressure 95/60, HR 110. Labs: total bilirubin 112, AST 320, ALT 98, ALP 200, INR 2.0, creatinine 88. Tbili 55. MELD score is 28, Maddrey Discriminant Function is 34. Ascitic fluid analysis shows PMN count 270 cells, culture pending. Abdo US show cirrhotic morphology but no biliary dilation. What is the next best step?

- A. Arrange for urgent ERCP - **No evidence for cholangitis or obstructive jaundice on imaging.**
- B. **Start IV Ceftriaxone + Albumin** – **Yes, PMN count in ascitic fluid ≥ 250 , which meets criteria for SBP; standard of care is IV CTX + albumin, to reduce risk of HRS and mortality risk.**
- C. Order TJ liver biopsy for confirmation of histologic diagnosis – **Diagnosis of Alc Hep is clinical (bx typically not required unless alternative diagnosis is being suspected such as AIH)**
- D. Start Prednisolone + IV N-Acetylcysteine for severe alcohol-associated hepatitis- **while this patient meets criteria for severe alc hep, the fever + SBP on ascites is a C/I to steroids. Priority is to treat the SBP**

Celiac Disease

Celiac disease is an immune-mediated enteropathy due to gluten hypersensitivity

- **Risk Factors:** Northern European descent, family history (1st degree relative), T1DM, AI disease, Down Syndrome (6x risk), Turner's syndrome, IgA deficiency
- **Symptoms & Signs:**
 - Diarrhea/ steatorrhea, abdo pain, bloating, weight loss, anemia, osteoporosis, enamel defects, elevated transaminases (mild), infertility
 - Vitamin & mineral deficiencies: A, D, E, B12, Fe, Ca
 - Dermatitis herpetiformis rash seen in 25% of pt with celiac
 - Tx = Gluten Free Diet (GFD), Dapsone (after r/o G6PD deficiency)
- **Diagnosis:**
 - Serology: Anti-TTG IgA & IgA level (to r/o IgA deficiency) +/- Anti-Deamidated Gliadin Peptide [DGP] IgG if ↓ IgA (anti-endomysial Ab [EMA] mentioned in guideline as high specificity but not widely available, and used moreso as confirmatory testing in kids who cannot undergo EGD with biopsy)
 - Histology: Upper endoscopy with duodenal biopsies (histologic changes as per Marsh Classification: including intraepithelial lymphocytosis, crypt hyperplasia, and villous atrophy)
 - EGD indicated if 1. symptomatic 2. elevated TTG-IgA 3. high pre-test suspicion of celiac >5% (even if negative serology)



DH: Grouped, pruritic papules and vesicles on extensor surfaces. Specific sign; >90% of individuals with DH have CD

Simplified Diagnostic Algorithm for Celiac Disease

*Screen for CD in:

- Irritable bowel syndrome
- T1DM with symptoms
- 1st degree relative with Celiac
- Elevated liver enzymes
- IDA/unexplained anemia
- Metabolic bone disease, premature osteoporosis
- Amenorrhea/infertility
- Down/Turner's Syndrome
- Dermatitis Herpetiformis

Adult patient with signs/symptoms of Celiac Disease (CD)

OR

***Individuals at risk for CD**

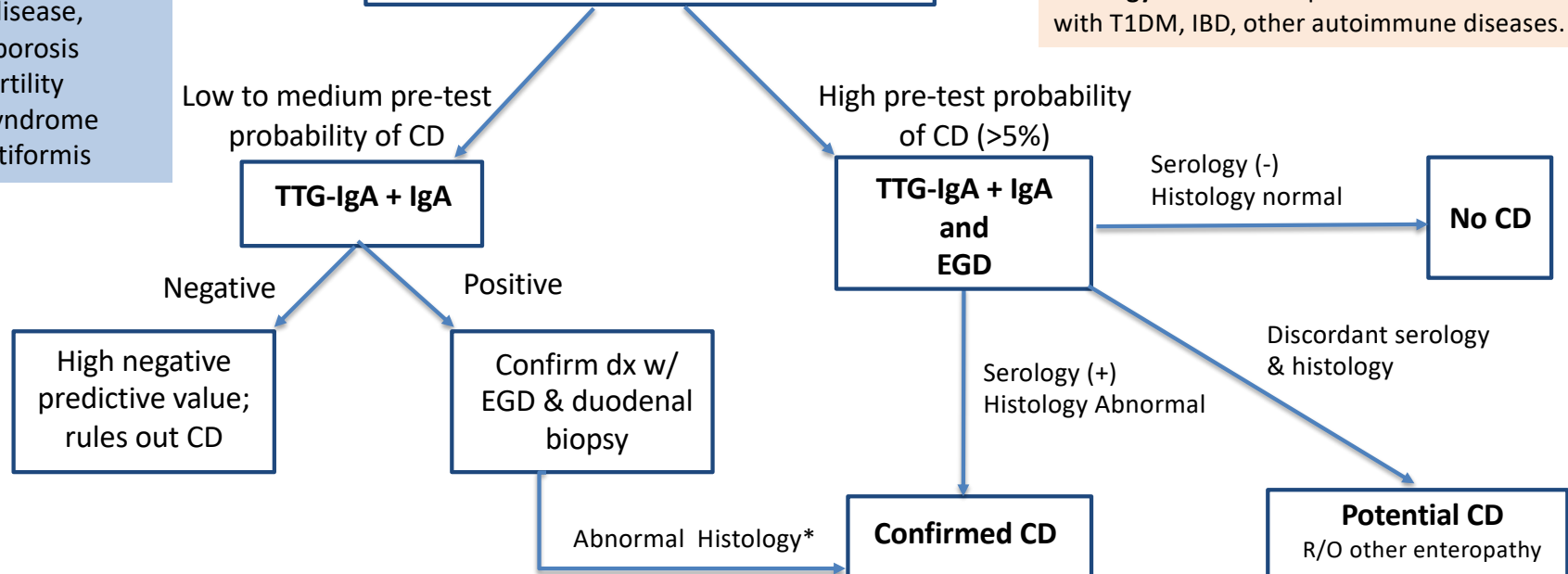
(assuming patient is eating gluten)

What if my patient has IgA deficiency?

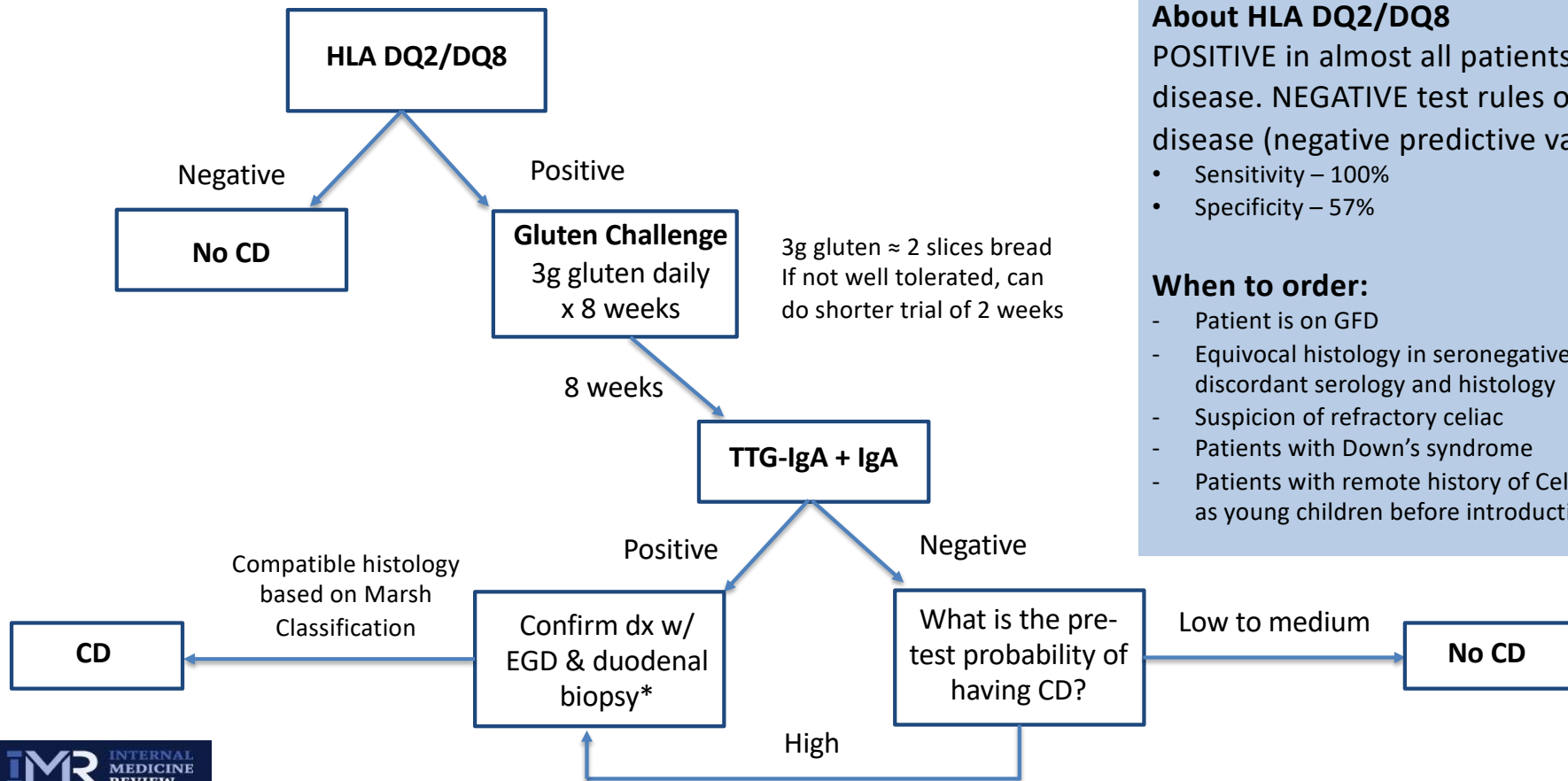
Consider Anti-DGP IgG or TTG-IgG as alternative serologic testing

Why EGD even with TTG-IgA?

Serology is not 100% specific – can see false + with T1DM, IBD, other autoimmune diseases.



What if my patient is on a Gluten-Free Diet (GFD)?



About HLA DQ2/DQ8

POSITIVE in almost all patients with Celiac disease. NEGATIVE test rules out Celiac disease (negative predictive value = 100%)

- Sensitivity – 100%
- Specificity – 57%

When to order:

- Patient is on GFD
- Equivocal histology in seronegative patients or discordant serology and histology
- Suspicion of refractory celiac
- Patients with Down's syndrome
- Patients with remote history of Celiac dx (especially as young children before introduction of TTG-IgA)

Celiac Disease

- **Treatment:** Lifelong gluten-free diet (GFD)
- **Follow-up:**
 - Ensure adherence to GFD
 - Diet history → consider dietitian consult
 - F/U serology at 6 & 12 months then annually to ensure ↓ antibody titres
 - Repeat EGD to assess for mucosal healing
 - Monitor for complications
 - Nutritional deficiencies and anemia (Fe, folate, Vit D, B12)
 - Osteopenia/osteoporosis (BMD at dx and then per guidelines)
 - Elevated liver enzymes
 - Dermatitis herpetiformis
 - Enteropathy-associated T cell lymphoma (EATL)
 - Consider refractory celiac disease and progression to EATL if a celiac patient stops responding to GFD → repeat EGD

Gluten Containing Foods – ‘BROW’

- **B**arley
- **R**ye
- **O**ats *often contaminated
- **W**heat

ACG 2023 Celiac Dx and Mgmt guidelines:

https://journals.lww.com/ajg/Fulltext/2023/01000/American_College_of_Gastroenterology_Guidelines.17.aspx?context=FeaturedArticles&collectionId=2

AGA 2019 guidelines: [https://www.gastrojournal.org/article/S0016-5085\(18\)35408-8/pdf](https://www.gastrojournal.org/article/S0016-5085(18)35408-8/pdf)

Question 2: IBD 2026

A 35F with a 10-year history of Crohn's ileocolitis presents with worsening perianal pain and drainage. She has been on vedolizumab for 18 months with stable luminal disease and no recent flares. On exam: HR 94, BP 118/72, T 37.8°C. Abdomen is soft, mild RLQ tenderness, but no peritonism. Rectal exam shows induration with external drainage. Labs: CRP 82 mg/L, Hgb 110 g/L. Colonoscopy 6 months ago showed only mild ileocolitis, unchanged. MRI pelvis today confirms a complex perianal fistula without abscess. Which of the following is the MOST appropriate next management step?

- A) Switch to methotrexate monotherapy
- B) Escalate to infliximab therapy, consider adding thiopurine
- C) Start corticosteroids for induction of remission
- D) Step up to Upadacitinib as immediate first-line for perianal Crohn's
- E) Stop biologics and manage conservatively with antibiotics only

Crohn's Disease

Inflammatory bowel disease characterised by transmural inflammation that can affect anywhere in the GI tract from 'gum to bum', often associated w/ development of abscesses, fistulas, and strictures.

- **Common disease locations:**

- Small bowel - terminal ileal involvement is most common
- Ileocolonic
- Colonic
- Mouth + upper GI tract

- **Clinical Manifestations:**

- Abdominal pain, diarrhea*, weight loss, fever, perianal symptoms (abscess, drainage from fistula)
 - Patients with stricturing CD, may present with obstructive symptoms (obstipation rather than diarrhea)*
- Subset of pts can have concurrent extra-intestinal manifestations: i.e. MSK- sacroiliitis, ankylosing spondylitis, erythema nodosum, pyoderma gangrenosum, episcleritis, PSC (more common in UC > CD)
- Labs: ↑ CRP, anemia, ↓ Albumin, ↓ Fe, ↓ B12, ↑ fecal calprotectin

- **Diagnosis:**

- Ileocolonoscopy & biopsy
 - Patchy inflammation, skip lesions, aphthous ulcers, cobble-stone mucosa
 - Rectal involvement + circumferential continuous inflammation less common in CD vs UC
- Small bowel imaging (CT/MR enterography, capsule endoscopy) +/- EGD if suspecting upper GI Crohn's
- No role for serology in diagnosis of IBD (e.g. ANCA/ ASCA)

Crohn's Treatment Options

Induction

- Mild
 - Budesonide (Entocort, Cortiment)
- Moderate to Severe
 - Budesonide
 - Prednisone/methylprednisolone
 - Methotrexate (MTX)
 - Anti-TNF – Infliximab (*Remicade*), Adalimumab (*Humira*)
 - Anti-Integrin – Vedolizumab (*Entyvio*)
 - Anti-IL12/23 – Ustekinumab (*Stelara*)
 - Anti-IL 23 – Risankizumab (*Skyrizi*)
*Mirikizumab (*Omvo*h), Guselkumab (*Tremfya*)
 - JAKi – Upadacitinib (*Rinvoq*)

Maintenance

- Mild
 - Thiopurine (AZP, 6-MP)
- Moderate to Severe
 - MTX
 - Thiopurine (AZP, 6-MP)
 - Anti-TNF – Infliximab (*Remicade*), Adalimumab (*Humira*)
 - Anti-integrin – Vedolizumab (*Entyvio*)
 - Anti-IL-12/23 – Ustekinumab (*Stelara*)
 - Anti-IL 23 – Risankizumab (*Skyrizi*), *Mirikizumab (*Omvo*h), Guselkumab (*Tremfya*)
 - JAKi – Upadacitinib (*Rinvoq*)

Thiopurines take ~8-12 weeks to have effect.
Refer for TPMT testing to identify those with low enzyme activity and at risk for adverse effects

*Mirikizumab & Guselkumab – **NEW** Health Canada Approved as of July 2025

MTX and Thiopurines – rarely used as monotherapy in mod-to-severe CD



IBD therapies- simplified summary

	Class	Best for	Avoid if
	Systemic steroids (bridge only)	Rapid induction in moderate–severe flares (CD & UC); taper once advanced therapy starts	Long-term use, brittle diabetes, osteoporosis, serious infection risk
	Budesonide	Mild–moderate ileocecal CD or UC induction	Need deep/rapid control; maintenance beyond short term
	5-ASA (mesalamine)	UC (mild–moderate) induction & maintenance	Renal disease (monitor)
"Biologics"	Anti-TNF	Severe, fistulizing/perianal CD ; fast induction in high-risk CD/UC	Recurrent serious infections, NYHA III/IV CHF, demyelinating disease, untreated TB/HBV
	Anti-integrin (vedolizumab)	UC & CD with great safety (older/comorbid pts); luminal disease – "Gut specific"	Need very rapid control; severe fistulizing disease
	Anti-IL-12/23 (ustekinumab)	CD & UC (luminal/penetrating), previous biologic-exposed (i.e. anti TNF failure)	Severe uncontrolled infection; rare hypersensitivity
	Anti-IL-23 (p19)	CD (risankizumab); UC (mirikizumab/guselkizumab where available); effective monotherapy	Same general biologic infection cautions
"Small-Molecules"	JAK inhibitors (tofa/upa/filgo*)	UC needing fast oral induction or bio-refractory; (upa also CD where approved)	High CV/VTE risk, pregnancy, recurrent zoster, active malignancy
	S1P modulators (ozani/etrasi)	UC oral option; steady maintenance once induced	Bradycardia/AV block, macular edema, uncontrolled HTN, significant drug interactions

CD: Treatment Patterns & Complications

General Principles

- **5-ASAs** are NOT effective for Crohn's ileitis, fistulizing or moderate-severe disease.
 - Sulfasalazine can be considered for mild colonic Crohn's disease but rarely used with expansion of biologics; occasionally used for concurrent treatment of Enteropathic arthritis
- **Surgery indicated for:** Intestinal obstruction, refractory/ fulminant disease, high-grade dysplasia/ cancer, severe perianal disease/ fistulas/ abscess, perforation.

Fistulizing Disease

- **Perianal fistulas:** Characterize with MR Pelvis (or EUS) + EUA (exam under anesthesia) → rule out abscess (*if abscess seen, tx with abx 1st +/- drainage before starting biologics)
- **1st line:** **Anti-TNFs** have the strongest evidence (IFX > ADA) ± immunomodulators
 - **Adjunctive:** ACG 2025 conditionally recommends using antibiotics with anti-TNF in peri-anal fistulas to improve clinical outcomes (Do NOT use abx alone to induce or maintain remission). For complex fistulas, consider surgical seton placement to allow drainage (often combined with Anti-TNF)
- Other options: Vedolizumab, Ustekinumab, Upadacitinib, Surgery

If a Crohn's patient has pain + fever

→ Order CT abdomen to look for abscess!

- If abscess ≥ 2 cm → refer to Surgery or IR for drainage AND TX with antibiotics
- *After* source control and infection treated → Biologics

Strictures

- **Cold stricture (fibrostenotic, no active inflammation)** → **conservative mgmt.** (bowel rest, NG tube), endoscopic balloon dilation, or surgery
- **Hot stricture** (active inflammation) → **steroid bridge** to maintenance therapy (**biologics**)

Ulcerative Colitis

Immune-mediated condition of the colon often associated with rectal inflammation, extending proximally to involve the adjacent colon in a continuous manner.

- **Disease Location:**
 - Proctitis – within 18 cm from the anal verge
 - Left-sided colitis – sigmoid to splenic flexure
 - Extensive/pancolitis – beyond the splenic flexure
- **Clinical Manifestations:**
 - ‘Proctitis’ – small volume + frequent BMs w/ blood, tenesmus, urgency, crampy abdominal pain
 - Fever, fatigue, weight loss
 - EIMs: similar to CD, but PSC incidence higher in UC
 - Inflammatory markers: Anemia, ↓ Albumin, ↑ ESR/CRP, ↑ fecal calprotectin
- **Diagnosis:**
 - Ileocolonoscopy + biopsies
 - Continuous inflammation from rectum proximally
 - Granular, friable mucosa
 - Bx – i.e. crypt abscesses, lamina propria cellularity

Truelove & Witts Criteria for Severe UC:

1. Bowel movements ≥ 6
2. Visible blood in stool
3. Pyrexia $T \geq 37.8^{\circ}\text{C}^*$
4. Pulse $> 90\text{bpm}^*$
5. Anemia $\text{Hgb} \leq 105^*$
6. ESR (or CRP) $>30^*$

Acute Severe UC (ASUC) when criteria for frequency of bowel movement and ≥ 1 features of systemic upset (*) are satisfied.

Ulcerative Colitis Treatment Options

• Induction

- Mild
 - 5ASA PO (extensive), PR (proctitis) (suppository <18cm, enema – to splenic flexure)
 - Budesonide
- Moderate to Severe
 - Budesonide
 - Prednisone/methylpred
 - Anti-TNF – Infliximab, Adalimumab, Golimumab
 - Anti-Integrin – Vedolizumab
 - Anti-IL-12/23 – Ustekinumab
 - Anti-IL 23 – Mirikizumab
 - JAK-inhibitor – Tofacitinib, Upadacitinib
 - Sphingosine 1-Phosphate Receptor Modulators – Ozanimod, Etrasimod

“Small molecules”, oral agents



• Maintenance

- Mild
 - 5ASA PO, PR
- Moderate to Severe
 - Thiopurine (AZP, 6-MP)
 - Anti-TNF – Infliximab, Adalimumab, Golimumab
 - Anti-Integrin – Vedolizumab
 - Anti-IL-12/23 – Ustekinumab
 - Anti-IL23 – Mirikizumab, Risankizumab
 - JAK-inhibitor – Tofacitinib, Upadacitinib
 - Sphingosine 1-phosphate receptor modulators – Ozanimod, Etrasimod

These lists are for reference - **know principles** (next slide), e.g. escalate therapy to include biologics / small molecules with worsening disease

IBD – General Principles

- **Work-up prior to starting advanced therapies (biologic or small molecule):**
 - HBV, HCV
 - TB skin test; If BCG vaccinated → CXR and/ or IGRA
 - Strongyloides serology if high pretest probability
 - TPMT testing if considering combination therapy with AZA/6MP
 - Beta-HCG → JAKi and S1PR modulators C/I in pregnancy
 - VZV titres for JAKi → vaccinate with Shingrix if not immune
 - Lipid profile for JAKi
 - Baseline ECG +/- ophthalmology assessment (if risk factors for retinal disease) for S1PR modulators
- **Treatment targets**
 - Clinical (symptoms, steroid free) remission
 - Biochemical (normalization of CRP and fecal calprotectin) remission, and
 - Endoscopic (mucosal healing) remission

IBD is associated with ↑ risk of colorectal cancer
See Med Onc lecture for more details

BONUS

Read on own

Screening: Serology +/- Stool O&P

Strongyloides Screening

Epidemiologic Risk category for Strongyloides exposure/infxn	Clinical Risk factor for disseminated Strongyloides	
	- HTLV-1 infxn - Pred $\geq 20\text{mg/d} \geq 2\text{wks}$ - Immunomodulatory agent (chemo, TNFa, biologics) - Hematologic Malignancy	No known defects in cell mediated immunity (healthy)
Birth/residence/longterm travel in SE Asia, Oceania, Sub-Saharan Africa, S. America, Caribbean	HIGH	Moderate
Birth/residence/longterm travel in Mediterranean countries, middle east, N Africa, Indian sub-continent, Asia	Moderate	Low
Birth/residence/longterm travel in Australia, North America or Western Europe	Overall VERY LOW risk [Florida, Kentucky, Virginia may be higher risk. Indigenous Australians higher risk]	

Travel = (6mos cumulative or more) w contact of skin with sand or soil, esp if exposure to rural/beach areas

Note: CCRIH recommends screening all refugees from SE Asia and Africa on entry to Canada. In the GI LECTURE we are talking about screening patients before immunomodulating therapy **

Boggild AK, CCDR 42(1);2016.



Inpatient IBD

IBD flares commonly overlap with infection

1. **Rule out C. diff**/ other infection if worsening diarrhea (**even if no recent antibiotics**)
2. Rule out abscess in a Crohn's patient with fever and abdominal/perianal pain
 - Use appropriate imaging (CT abdo, MR pelvis, EUS, etc.)
3. Withhold immunosuppression until (1+2) are addressed.
 - If the patient looks septic, cover with antibiotics
4. **CMV colitis** can co-exist with IBD and requires colonic biopsies to diagnose (CMV blood PCR is not enough; look for "owl-eye inclusion bodies" on pathology and CMV+ immunohistochem off biopsy)

Don't stop DVT prophylaxis even if patient having bloody diarrhea

- Risk of thrombosis > risk of life-threatening hemorrhage in hospitalized IBD patients with acute flare

Bile-salt diarrhea

- Occurs with ileitis or after ileal resection
- ALWAYS assess for and treat active IBD (CT imaging, CRP, Fecal calprotectin, colonoscopy) before treating for bile salt diarrhea
 - Treatment = cholestyramine (bile acid sequestrant)

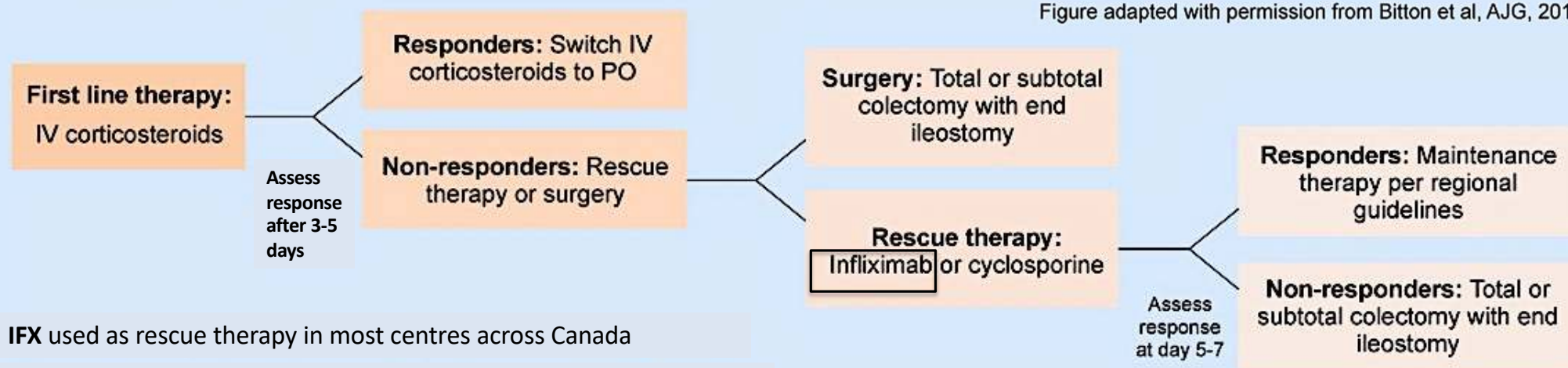


Inpatient IBD

- **Work-up:**
 - Labs – CBC, lytes, extended lytes, Cr, AST/ALT/ALP, bilirubin, INR, albumin, ESR/CRP
 - Stool tests – Stool C&S, Stool O&P, C. diff toxin, +/- fecal calprotectin
 - Imaging – AXR, +/- CT abdomen, +/- CT/MR enterography (i.e. small bowel imaging)
 - Endoscopy – for diagnosis, to assess disease severity, rule out mimics (ischemia, CMV), can send aspirates for micro
- **Management:**
 - IV fluids, bowel rest, VTE prophylaxis
 - Minimize narcotics, NSAIDS
 - Consider IV steroids once infection is ruled out
 - If minimal response after ~ 72h IV steroids → Infliximab
 - Consider general surgical consultation for treatment refractory colitis, toxic megacolon, perianal/intra-abdominal abscess, complex fistulizing Crohn's

Acute Severe UC (ASUC) Management

Figure adapted with permission from Bitton et al, AJG, 2012.



IFX used as rescue therapy in most centres across Canada

Response = < 4BMs/day x 2 days AND no rectal bleeding

Updated ACG 2025 Guidelines – emphasize **ruling out C diff (with stool assay), and CMV colitis with endoscopy** ideally within 24h-72h after presentation

- Multidisciplinary care with early surgical consultation is recommended
- **Indications for colectomy:** Toxic megacolon*, colonic perforation, severe refractory hemorrhage, refractory to medical therapy
 - In biologic era, rates of colectomy for ASUC have declined

*Radiographic megacolon (TC diameter >6cm)

At least 3 of

- Fever (>38 deg C)
- HR >120
- Neutrophils >10.5
- Anemia

At least 1 of

- Dehydration
- Altered sensorium
- Electrolyte disturbances
- Hypotension

Question 2: IBD – NEW 2026

A 35F with a 10-year history of Crohn's ileocolitis presents with worsening perianal pain and drainage. She has been on vedolizumab for 18 months with stable luminal disease and no recent flares. On exam: HR 94, BP 118/72, T 37.8°C. Abdomen is soft, mild RLQ tenderness, but no peritonism. Rectal exam shows induration with external drainage. Labs: CRP 82 mg/L, Hgb 110 g/L. Colonoscopy 6 months ago showed only mild ileocolitis, unchanged. MRI pelvis today confirms a complex perianal fistula without abscess. Which of the following is the MOST appropriate next management step?

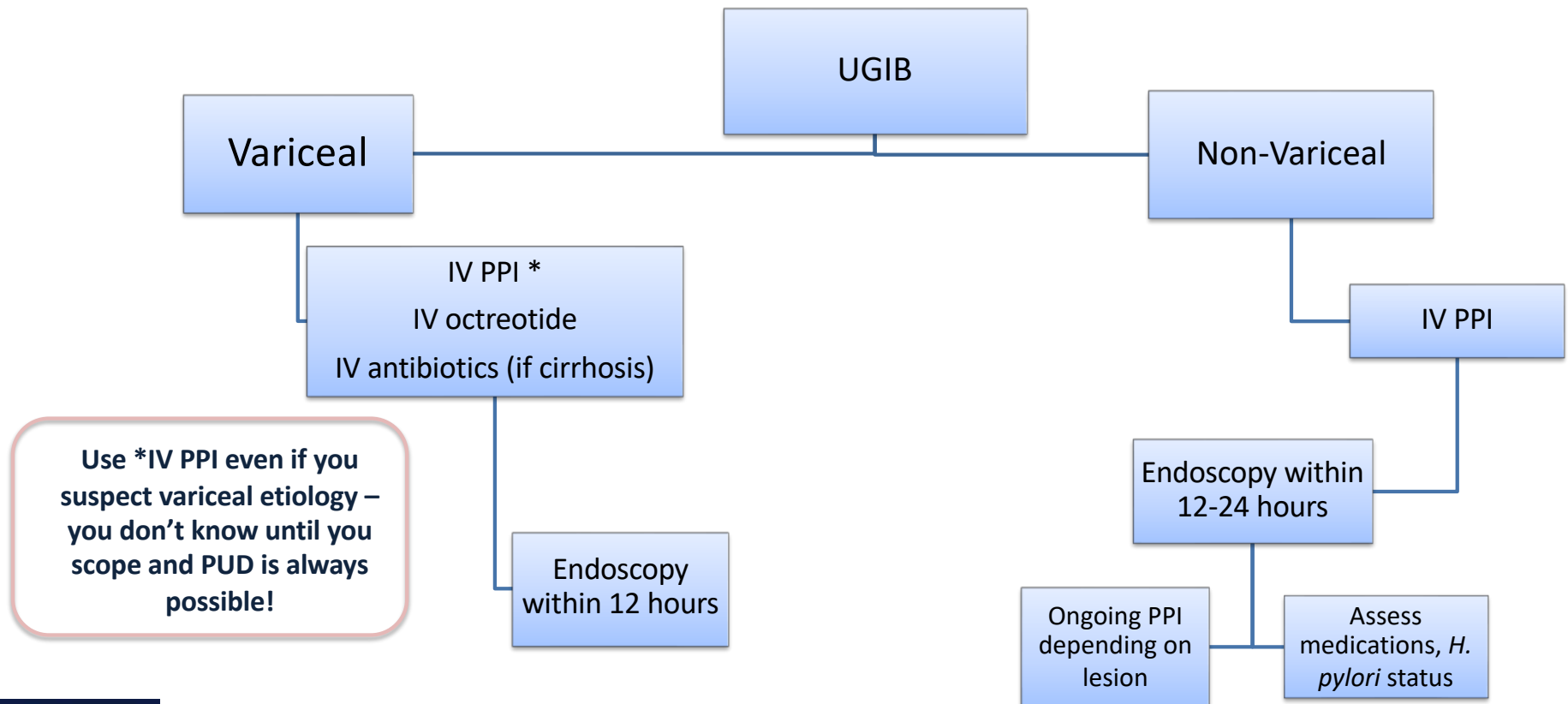
- A) Switch to methotrexate monotherapy **No, MTX monotherapy has no proven efficacy for moderate to severe CD and not indicated for peri-anal disease**
- B) Escalate to infliximab therapy, consider adding thiopurine **Yes, anti TNFs (IFX >ADA) remain 1st line in fistulizing perianal CD; thiopurines help reduce immunogenicity**
- C) Start corticosteroids for induction of remission **Steroids can help luminal flares, but they do not heal fistulas (in fact can worsen infection risk). In this case, luminal symptoms have remained stable**
- D) Step up to Upadacitinib as immediate first-line for perianal Crohn's **JAKi are only conditionally recommended in peri-anal disease after anti-TNF non-response/failure. Anti-TNFs remain first line as per latest ACG 2025 guidelines.**
- E) Stop biologics and manage conservatively with antibiotics only **While latest guidelines recommend Anti TNF + Abx for fistula, the role of antibiotics are adjuncts and will not treat the fistulas on their own**

Question 3 – UGIB (NEW 2026)

A 72-year-old man is admitted with melena. He is hemodynamically stable after fluids (BP 118/68, HR 86). Hb 98 g/L (baseline 128). He takes ASA 81 mg daily for secondary cardiovascular prevention and a PPI. Urgent endoscopy is planned today. What is the best next step regarding antiplatelet management?

- A. Stop ASA now and restart 7 days after endoscopic hemostasis
- B. Continue ASA without interruption
- C. Stop ASA and give a platelet transfusion to reverse its effect
- D. Hold ASA now; restart on the day of hospital discharge
- E. Give desmopressin to reverse antiplatelet effect

Upper GI Bleeding



UGIB

- **Pre-endoscopy (**POSSIBLE ORAL SCENARIO**)**

- ABC's, monitoring, O₂, 2-large bore IV's, fluids (ABC-MOIF)
- CBC, urea, Cr, cross and type, INR/PTT, Liver enzyme and function tests
 - Use Glasgow-Blatchford score to identify low-risk patients that can be discharged with outpatient follow up [ACG 2021]
 - Uses Hgb, BUN, SBP, HR, presence of melena/syncope/CHF/liver disease. Score 0-1 (/23) = very low risk → outpatient workup appropriate
- Transfuse
 - **Hb > 70**; consider higher target (>80) if symptomatic/ active cardiac ischemia, or unstable
 - **Plts > 50** if actively bleeding/unstable
- Reverse anticoagulation
 - See slide on antithrombotic mgmt. in GI bleed
- PPI therapy (IV bolus + infusion or BID dosing)

Tranexamic Acid **does not** reduce mortality in UGI Bleed based on HALT-IT Trial (Lancet 2020) – may ↑thrombosis and seizures, do not use routinely!

? 🤔 Fibrin clots form best *at neutral pH*. Hence TXA blocking fibrinolysis not terribly effective with gastric pH (<4 = reduced fibrin clots!), and PPI works as it raises gastric pH.

UGIB in Patient with Cirrhosis

UGIB (variceal or not) + Cirrhosis (or suspected non-cirrhotic portal HTN)

1. **Restrictive transfusion strategy** (Threshold <70 g/dL)
 - No FFP
 2. **IV vasoactive drug** (octreotide, somatostatin, terlipressin [not yet in Canada] or vasopressin) as soon as variceal bleed is suspected; continue x 2-5 days post-endoscopy if variceal bleed
 3. **IV PPI** -if varices treated, can stop ; continue if indicated for non-variceal ulcer bleeding
 4. **Endoscopy within 12h** of presentation
 5. **Antibiotic prophylaxis** is indicated for any GI bleed in a patient with cirrhosis
 - Ceftriaxone 1g/24h x 5 days
- For refractory variceal bleed → rescue TIPS
 - Following variceal bleed + EVL:
 - Initiate NSBB when vasoactive drugs stopped
 - NSBB not required if pt had a TIPS placed; TIPS is rescue if bleed despite NSBB+EVL

Pre-emptive TIPS can be considered in high-risk patients (Child Pugh C cirrhosis or Child Pugh B 8-9 with active bleeding at time of EGD) within 72h of AVH .

Diaz-Soto MP, Garcia-Tsao G. *Therap Adv Gastroenterol.* 2022;15:17562848221101712

For exam: prioritize ABCs, medical Tx with abx/PPI/vasoactive therapy, and GI consult for EGD within 12h for suspected variceal bleed!

Non-variceal UGIB

- Endoscopic management
 - Erythromycin 250 mg IV suggested 30 min before endoscopy
 - Endoscopy should be performed within 24 hours (or within 12 hours in high-risk patients, including suspected variceal bleeds)
 - Endoscopic therapy only for high-risk ulcers (Forest IA, IB, IIA ulcers)
- If ongoing/ re-bleeding: repeat EGD, consider IR embolization or surgical management
- Consider radiation for tumour bleeds

Barkun et al. Management of Nonvariceal Upper Gastrointestinal Bleeding: Guidelines Recommendations from the International Consensus Group. *Annals of Internal Medicine*. Dec. 2019.

ACG Guideline: Upper Gastrointestinal and Ulcer Bleeding. *Am J Gastroenterol*. 2021 May 1;116(5):899-917. doi: 10.14309/ajg.0000000000001245.

Non-variceal UGIB

- **Post-endoscopy**

- High-risk ulcers → IV PPI BID or PPI infusion x 72 hours, then oral PPI BID for at least the first 2 weeks after endoscopy
- Low-risk ulcers → Oral PPI (no guidelines; typically BID x 2-4 weeks then stop)
- **Restart anti-platelets and anticoagulation as soon as possible** (see ACG-CAG guidelines next slide)
 - Restarting OAC and/or antiplatelets not associated with increased GI bleed risk but IS associated with a reduction in all cause mortality (*Witt DM Hematology ASH education Program 2016:620*).
 - Timing – variable decision incorporating patient factors and endoscopic findings (Forrest class)

- **Consider risk factors**

- Test all with PUD for *H. pylori* – can biopsy during EGD or check serology, treat if positive
- Stop all unnecessary NSAIDs, stop ASA for primary prophylaxis
- For CV patients requiring ongoing ASA and/or clopidogrel, consider indefinite PPI prophylaxis
- Indefinite PPI if unclear cause of PUD (conditional recommendation)

Anti-thrombotic management

(Joint ACG-CAG Guideline April 2022)



Acute GI Bleeding – All guidance “suggest” due to low level of evidence → use your clinical judgement!

i) Patient stable, “hospitalized or under observation for GI Bleed [upper or lower]”

- **ASA 81-325mg for 2^o cardiovascular prevention** → Suggest no interruption; if interrupted → re-start on the day that hemostasis is endoscopically confirmed.
- **Anticoagulation reversal?**
 - Warfarin → Suggest AGAINST vitamin K and FFP use. No recommendation for/ against PCC.
 - Do not routinely give Vitamin K unless low dose to correct supratherapeutic INR
 - DOAC → Suggest AGAINST reversal with PCC/ idarucizumab/ andaxnet alpha

ii) Life threatening hemorrhage, “major clinically overt or apparent bleeding, resulting in hypovolemic shock, or severe hypotension requiring pressors or surgery”, ↓Hgb >50, transfusion ≥5u PRBC, or causing death”

- Warfarin → PCC “could be considered” (or in patients in whom massive transfusion is undesirable)
 - Uncertain re use of vitamin K – see above, *conditional recommendation* to not use if INR in target range and intervention planned
- DOAC → selective use of PCC (for anti-Xa DOAC) or idarucizumab (for Dabigatran) for those who took DOAC within 24h who have life threatening bleed “may be appropriate”
- No Platelet transfusions to reverse effect of antiplatelet!

BONUS
Read on own

Anti-thrombotic management peri-procedure (Joint ACG-CAG Guideline April 2022)

Anti-thrombotic management in the **elective** endoscopy setting*:

- **ASA 81-325 mg for 2° prevention** → Suggest no interruption
- **DAPT** → Suggest interrupting only the P2Y₁₂ inhibitor; Re-start 0-7d post-endoscopy (no consensus)
- **P2Y₁₂ monotherapy** → No consensus for or against interruption*, or when to restart
- **DOACs** → Suggest interruption[^]; Re-start 0-7d post-endoscopy (no consensus)
- **Warfarin** → Suggest no interruption[#], and if interrupted → Suggest no need to bridge

*Individualize risk of thrombosis versus bleeding in setting of high risk endoscopic procedures:

(eg) high risk examples: >1cm polypectomy, ERCP w sphincterotomy, PEG/PEJ tube, ampullary resection, treatment of varices including band ligation ...

[^] In line w thrombosis Canada – see Perioperative Medicine Lecture

In an Oral scenario “I will liaise with my GI/Gen Surg colleague and their heme/cardio”...

[#]Based on a single observational study, depends on what you’re expecting to find on endoscopy

ACG-CAG Guideline April 2022: Management of Anticoagulants and Antiplatelets During Acute Gastrointestinal Bleeding and the Periendoscopic Period
https://journals.lww.com/ajg/Fulltext/2022/04000/American_College_of_Gastroenterology_Canadian.16.aspx

Question 3 – UGIB (NEW 2026)

A 72-year-old man is admitted with melena. He is hemodynamically stable after fluids (BP 118/68, HR 86). Hb 98 g/L (baseline 128). He takes ASA 81 mg daily for secondary cardiovascular prevention and a PPI. Urgent endoscopy is planned today. What is the best next step regarding antiplatelet management?

- A. Stop ASA now and restart 7 days after endoscopic hemostasis **No. A prolonged interruption increases major adverse cardiovascular events without clear bleeding benefit in a stable patient.**
- B. Continue ASA without interruption** **Correct.** In stable GI bleeding, guidelines suggest no interruption of low-dose ASA used for secondary prevention.
- C. Stop ASA and give a platelet transfusion to reverse its effect **No. Transfusing platelets does not meaningfully reverse aspirin's effect in GI bleeding and may cause harm.**
- D. Hold ASA now; restart on the day of hospital discharge **No need to hold ASA and delaying until discharge leaves him unnecessarily exposed to thrombotic risk; if it is held (in the event of unstable GI bleed) should restart after confirmed hemostasis.**
- E. Give desmopressin to reverse antiplatelet effect **Not recommended for routine ASA effect in GI bleeding; evidence is limited**

H. pylori Fast Facts ACG 2024

– H. Pylori ↑ risk of gastric cancer

- ACG 2024 supports a test-and-treat approach → if pt has indication for testing, they should be offered HP tx if test positive and receive test-of-cure thereafter

– Who to test?

- Any patient with peptic ulcer disease (prior history or current)
- MALT lymphoma → *tx of HP alone will cure most cases!*
- Uninvestigated or functional dyspepsia → HP treatment benefits are modest but only potential curative tx for FD!
- Long-term NSAIDs/ASA
- Unexplained Fe deficiency anemia
- ITP

Given the new indications in the ACG guidelines are not yet adopted in Canada, focus on these for your exam!

NEW ACG 2024 Adult household members of individuals who have a + non-serologic test for HP

NEW ACG 2024 Primary & secondary gastric adenocarcinoma prevention in patients with:

- Current or prior hx of gastric adenocarcinoma or early gastric cancer resection
- 1st degree relative w/ gastric cancer
- Gastric pre-malignant conditions including atrophic gastritis, intestinal metaplasia, dysplasia
- Autoimmune gastritis
- Gastric polyps (adenomas or hyperplastic polyps)
- Immigrants from high gastric cancer incidence regions

ACG Guideline: Treatment of *H. pylori* Infection (*Am J Gastroenterol* 2024; 119(9):p 1730-1753.

CAG Guideline: The Toronto Consensus for the Treatment of *Helicobacter pylori* in adults (*Gastroenterology*. 2016 Jul;151(1):51-69.e14.)

H. pylori Diagnosis

– Testing to detect *active* infection

- Stool Antigen (SN 94% / SP 92%)
- Histology (SN 96% / SP 99%)
- Biopsy Culture (SN 88-95% / SP 95-100%)
- Urea Breath Test (SN 90-95% / SP 95-100%)

– Testing to detect *current or prior* infection

- Serology i.e. serum IgG (SN 85% / SP 80%)
 - If negative in a patient with bleeding ulcer, repeat testing is recommended when acute bleeding episode resolves

Note for New ACG Guidelines
Re: populations targeted for
Primary and Secondary
Prevention of Gastric CA:

“H pylori treatment should
be offered to all patients
with active H pylori
infection is indicated by a
positive NON
SEROLOGICAL TEST”

[e.g. urease breath, biopsy
or stool Ag]

H. pylori Treatment

2016 Toronto Consensus Guidelines & NEW 2024 ACG :

- Triple therapy no longer recommended unless local resistance to clarithromycin < 15%
- **1st Line Treatment-Naïve Patients**
 - ***PBMT (PPI/Bismuth/Metronidazole/Tetracycline) x 14 days***
 - Supported by both CAG & ACG 2024 – go w/ this for exam answer if have to choose between PBMT & PAMC!
 - **PAMC x 14 days** – Recommended as 1st line in Toronto Consensus but not supported by recent ACG 2024 given increasing rates of clarithromycin and amoxicillin resistance in N. America. Practically still used in Canada!
- **Treatment Failure:**
 - **PBMT x 14 days**
 - Preferred treatment if patient has never received optimized bismuth therapy
 - **PAL (PPI/Amoxicillin or metronidazole if pen allergy)/Levofloxacin) x 14 days**
 - Preferred in those previously treated with bismuth or rifabutin based therapy
 - **PAR (PPI/Amoxicillin/Rifabutin) x 14 days**
 - Toronto Consensus states this should be reserved for those who have failed ≥ 3 tx
 - ACG 2024 lists this as both 1st line tx option (conditional recommendation based on low quality of evidence) and 2nd line in those with tx failure
- Confirm eradication in **all** treated patients
 - Urea breath test, stool antigen test, or repeat gastric biopsy

Wait ≥ 4 weeks after completing antibiotic therapy and $\geq 1-2$ weeks after PPI therapy before testing for *H. pylori* to improve test accuracy!

Management of GERD – ACG 2022 Guideline

DIAGNOSIS OF GERD

Classic symptoms (heartburn, regurgitation, PPI-responsive):

→ **Clinical diagnosis** = no need for EGD or pH monitoring.

EGD to assess for esophagitis or Barrett's only if:

- *atypical symptoms, red flags, or refractory to PPI:**

→ **24h pH monitoring** if EGD is non-diagnostic.

✗ Do not use barium studies or manometry to diagnose GERD.

*Red flags = dysphagia, weight loss, GI bleeding, anemia

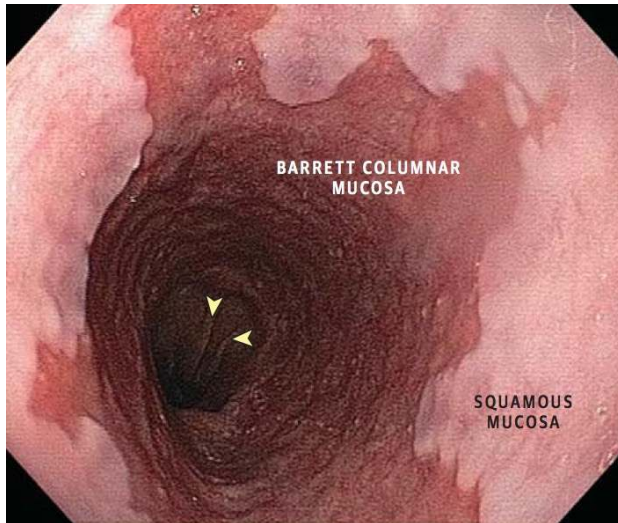
- **Management of GERD** (in absence of red flag* symptoms)
 - Non pharm mgmt. = weight loss, avoid trigger foods, quit smoking, elevate HOB, etc)
 - Trial PPI once daily 30-60min before a meal x 8 weeks
 - Symptoms improve? → Try to wean PPI, if unable: maintain lowest dose PPI
 - Symptoms refractory? → Ensure optimization of PPI therapy → Sx refractory despite optimization → EGD off PPI, if normal → 24h pH monitoring → if negative, discontinue PPI, consider other causes
 - Consider sucralfate in pregnancy, H2R blocker at night for nocturnal symptoms
 - Recommend *AGAINST* prokinetics, sucralfate, baclofen, routine H2RA in PPI non-responders
- **Extra-esophageal symptoms** (cough, asthma, laryngitis, dental erosions)
 - Rule out non-GI causes (e.g. PFTs, referral to resp/ENT/dentist)
 - Mainstay of diagnosis = 24h pH monitor
 - Consider BID PPI x 8-12 weeks if pt also has typical GERD symptoms; failing this → EGD off PPI

Anti-Reflux Surgery

- **Candidates:** Confirmed GERD (EGD or pH) with normal manometry, persistent symptoms or large hiatal hernia, or severe esophagitis
- **Options:** Nissen fundoplication, magnetic sphincter, Roux-en-Y (if obese)
- **Alternatives:** Endoscopic incisionless fundoplication if surgery declined

ACG Clinical Guideline for the Diagnosis and Management of Gastroesophageal Reflux Disease. The American Journal of Gastroenterology 117(1):p 27-56, January 2022.

Barrett's Esophagus



ACG Guideline: Diagnosis and Management of Barrett's Esophagus (*Am J Gastroenterol* 2022; 117 (4): 559-587).

NEW! AGA Clinical Practice Guideline on Endoscopic Eradication Therapy of Barrett's Esophagus and Related Neoplasia. *Gastroenterology*. 2024 Jun;166(6):1020-1055

- **Diagnosis:**
 - Metaplasia of normal squamous epithelium to salmon-colored columnar epithelium ≥ 1 cm proximal to the gastroesophageal junction.
 - Biopsy confirmation of columnar intestinal metaplasia, +/- goblet cells.
 - Take multiple (≥ 8) biopsies if suspected by typical endoscopic appearance!
- **Malignant potential:**
 - 30 fold $\uparrow$ risk of esophageal adenocarcinoma vs average population
 - Low overall annual risk (0.1% – 3%)
 - Dysplastic > Non-dysplastic; long segment > short segment (<3cm)
- **Risk Factors:**
 1. Chronic GERD (weekly symptoms for ≥ 5 years)
 2. Age > 50
 3. Male gender
 4. Tobacco use
 5. Central obesity
 6. Caucasian
 7. Family history of Barrett's esophagus/ esophageal adenocarcinoma

Alcohol is **not** known to $\uparrow$ risk of Barrett's or esophageal adenocarcinoma! Only SCC

Screening for Barrett's Esophagus

ACG April 2022 Guidelines

(strength of recommendation: **conditional**; quality of evidence: **very low**):

- Single screening endoscopy (or capsule + biomarker) for patients with chronic GERD (weekly symptoms for ≥ 5 years) and ≥ 3 risk factors for Barrett's including:
 - Age > 50
 - Male
 - Caucasian
 - Obesity
 - Tobacco smoking
 - 1st degree relative with Barrett's or esophageal adenocarcinoma.

Canadian Task Force 2020:

- *"We recommend not screening adults (≥ 18 years) with chronic GERD for esophageal adenocarcinoma or precursor conditions (Barrett esophagus or dysplasia) (strong recommendation; very low-certainty evidence)"*



Go with this for your exam, considering ACG was a conditional recommendation re screening

Management of Barrett's Esophagus

(ACG 2022 & NEW AGA 2024 Guidelines)

1. **NO METAPLASIA** → Repeat EGD w/ biopsy in 1-2 yrs to R/A suspicious area

2. **INTESTINAL METAPLASIA** → Confirms BE; assess if there is any dysplasia

- **Non-Dysplastic BE (NDBE)**

- EGD q3y if long-segment BE (≥ 3 cm)
- EGD q5y if short-segment BE (< 3 cm)

- **Indefinite Dysplasia**

- Review biopsies with 2nd GI pathologist
- PPI BID → repeat EGD w/ biopsy in 6 months

- **Dysplasia**

- Review biopsies w/ 2nd GI pathologist
- **LGD** → EET preferred over surveillance in AGA 2024 followed by EGD to reassess at 1y and 3y then q3-5y as per NDBE surveillance guidelines (note from ACG recommendation of 1y then 2y screen)
- **HGD** → EET → EGD at 3 mo, 6 mo, 12 mo, then q1y w/ biopsies

Submucosal T1b cancer → refer to thoracics for esophagectomy (can consider EET in select cases with low-risk features)

LGD = Low grade dysplasia

HGD = high grade dysplasia

EET = endoscopic eradication therapy

AGA 2024 recommends the following in **all** patients with bx confirmed BE:

- Tobacco cessation
- Weight loss if overweight
- Acid control – **once daily PPI**
 - Exceptions: refractory GERD symptoms or undergoing EET

Endoscopic Eradication Therapy:

Includes various resection techniques (i.e. EMR & ESD) of visible lesions and ablative techniques (e.g. radiofrequency ablation) of IM that are performed q 2-3mo with goal of achieving complete eradication of intestinal metaplasia & neoplasia (CEIN)

Approach to Dysphagia:

(1) History and Physical

1. Where is it?

- **Oropharyngeal** = Trouble initiating swallow, choking, drooling, symptoms **above** sternal notch, **neuro** history (e.g. stroke, Parkinson's)
- **Esophageal** = Food “stuck” **below** sternal notch or regurgitation

2. What kind of food?

- **Solids only** → **Structural** (e.g. stricture, ring, tumor) – may progress to solids + liquids
- **Solids + liquids** → **Motility** (e.g. achalasia, spasm)

3. How often?

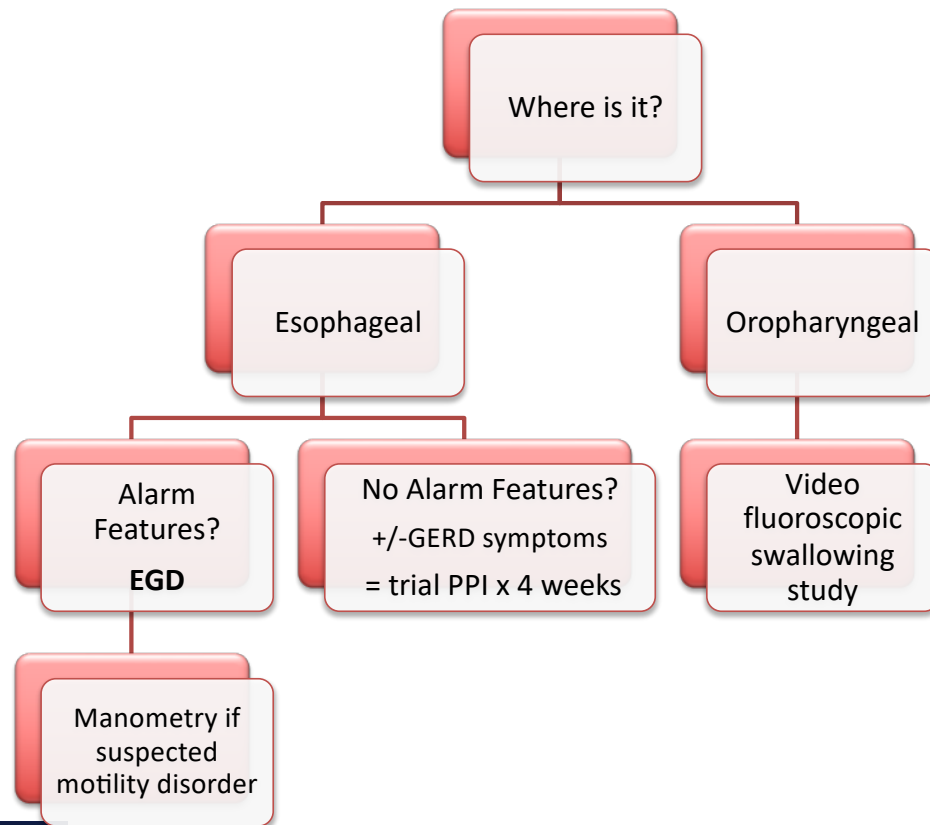
- **Intermittent** → Motility disorders or esophageal (eg) Schatzki ring
- **Progressive** → Stricture, cancer, achalasia

4. Red flags or Alarm Features?

-  Age >50, weight loss, odynophagia, vomiting, GI bleeding

Approach to Dysphagia:

(2) EGD +/- Imaging



Key Concepts from CAG 2018 Guideline

- EGD is recommended over barium swallow if esophageal dysphagia.
 - Can biopsy for eosinophilic esophagitis on endoscopy
- High resolution esophageal manometry is recommended over barium swallow for investigation of esophageal motility disorders
 - EGD first to rule out pseudoachalasia

Achalasia:

Motility disorder causing incomplete relaxation of lower esophageal sphincter (LES) & ineffective esophageal peristalsis

Key Features:

- Esophageal dysphagia to **solids & liquids** (progressive)
- **Regurgitation**, chest pain, weight loss
- **PPI-refractory** symptoms
- **Epidemiology:** Rare (~10 per 100,000), M = F, age **30–60**
- **Complications:** Megaesophagus, sigmoid esophagus, ↑ risk of **esophageal SQUAMOUS cell carcinoma**

Diagnosis:

- 1) **EGD (rule out pseudoachalasia (tumour/obstruction) then**
- 2) **High resolution manometry** (Gold standard test for achalasia)
 - Measures pressures & peristaltic waveforms
 - Impaired relaxation of LES and abnormal peristalsis are diagnostic
 - Barium Swallow not required but often done due to availability/access to manometry (“Bird’s beak esophagus”)



Management: (ACG 2020 Guideline)

- **Poor surgical candidates due to comorbidities:**
 - Endoscopic Botox Injection – 1st line
 - Smooth muscle relaxants* – CCB, nitrates
 - Ex. Nifedipine 10-30mg SL TID, ISDN 5mg SL TID.
- *Least effective therapy
- **Good surgical candidate:**
 - Pneumatic dilatation
 - Risk of tear/perforation
 - Laparoscopic Heller Myotomy
 - Often performed with fundoplication (anti-reflux surgery)
 - Peroral endoscopic myotomy (POEM)
 - Often complicated by GERD – patients need lifelong acid suppression
 - Esophagectomy – *last resort*
 - For patients with megaesophagus, sigmoid esophagus, failure of above treatments

What about CHAGAS? Esophageal involvement from chagas is indistinguishable from achalasia on endoscopy! It is diagnosed via serology. Biopsies are not helpful! Treatment of Chagas esophagus is the same.
Chagas is rare outside of the southern S. America eg Bolivia, Uruguay, Brazil

Eosinophilic Esophagitis (EoE)

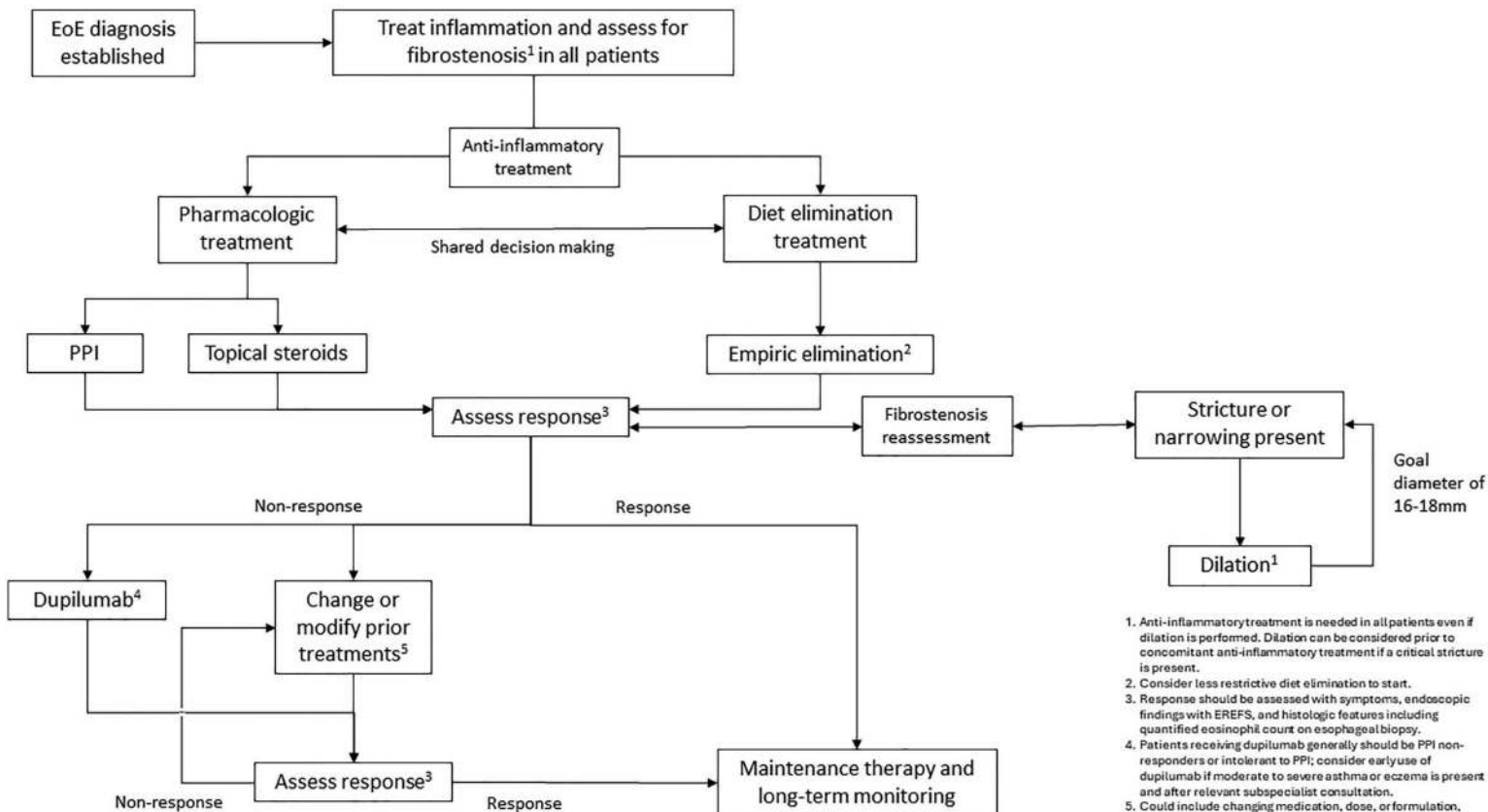
- **Diagnosis:** Immune-mediated disease characterized by: **(1) Sxs of Esophageal dysfunction** (i.e. dysphagia, food bolus impaction, chest pain, heartburn, failure to thrive in children), **(2) > 15 eos/hpf** on esophageal biopsy **(3)** ruled out other causes of esophageal eosinophilia*
- **Risk factors:**
 - M > F (3:1). Peak incidence at 30-44 yo.
 - History of atopy (asthma, eczema, allergic rhinitis), food allergies, autoimmune disease
- **Endoscopic findings: Use EREFS Score:** Edema, Rings (i.e. classic 'Trachealization' of the esophagus aka 'Feline Esophagus'), Exudates, Furrows, Strictures
- **Treatment goal = Symptomatic , endoscopic, and histologic improvement**
 - Non-pharmacologic: ACG 2025 recommends stepwise elimination approach – i.e. start off with 2FED (dairy + wheat), reassess with endoscopic + bx then escalate to 4FED (egg+ soy) if needed up to →6FED 6 food elimination diet (eggs, soy, animal dairy, wheat, tree nuts, seafood).
 - Pharmacologic :
 - **1st line:** PPI (bid dosing recommended to start) vs. topical steroids (orodispersible budesonide [Jorveza] preferred > swallowed fluticasone/budesonide)
 - **2nd line: Dupilumab** (anti-IL 4/IL13) – FDA approved 2024; for patients not responsive to PPIs – may be considered earlier in select patients with concomitant atopic conditions
 - Endoscopic dilation for symptomatic strictures



***Secondary causes of esophageal eosinophilia:**
GERD, achalasia, connective tissue diseases, Crohn's hypermobility syndromes, pill esophagitis, pemphigus, hyperIgE syndrome, GVHD

BONUS
Read on own

ACG 2025 EoE Treatment algorithm



- Allergy testing is NOT recommended for w/u of EOE (IgE and skin prick tests have poor predictive value)
- ACG suggests against omalizumab, cromolyn, and montelukast

1. Anti-inflammatory treatment is needed in all patients even if dilation is performed. Dilation can be considered prior to concomitant anti-inflammatory treatment if a critical stricture is present.
2. Consider less restrictive diet elimination to start.
3. Response should be assessed with symptoms, endoscopic findings with EREFS, and histologic features including quantified eosinophil count on esophageal biopsy.
4. Patients receiving dupilumab generally should be PPI non-responders or intolerant to PPI; consider early use of dupilumab if moderate to severe asthma or eczema is present and after relevant subspecialist consultation.
5. Could include changing medication, dose, or formulation, moving to a more restrictive diet, or considering a clinical trial.

Acute Pancreatitis

Definition (need 2/3 criteria for diagnosis):

1. Abdominal pain consistent with disease
= acute severe epigastric/ LUQ, radiates to back, better leaning forward
2. Lipase and/or amylase $\geq 3\times$ ULN
3. Characteristic findings on imaging

Etiology:

#1: Gall stones - always order **U/S** to rule out gallstones (Endoscopic ultrasound [EUS] or MRCP if high clinical suspicion and U/S negative or in idiopathic pancreatitis)

#2: EtOH

#3: Other – Triglycerides (> 11 mmol/L), hypercalcemia, drugs (GLP1 agonists, 5-ASA, thiazides, azathioprine), autoimmune, **post-ERCP**, trauma, viruses, **malignancy** (suspect in pt > 40 yo), hereditary, smoking, scorpion bite, vasculitis, idiopathic

Management of Acute Pancreatitis: (ACG 2024)

Severity: *Presence of organ failure or pancreatic necrosis defines severe pancreatitis.*

- ACG 2024: Assess early signs/risk factors of severe disease rather than depending on a single scoring system for prognostication.
 - Altered mental status
 - Imaging: Pulmonary infiltrates or pleural effusions or multiple extra-pancreatic collections
 - SIRS (2 or more of HR>90, RR >20, T >38 or <36, WBC <4 or >12)
 - Labs – rising BUN, hematocrit (HCT), Cr
 - Demographic risk: age >55, obesity (BMI>30), comorbid disease
- CT scan **NOT** required for diagnosis or prognosis (consider if failure to improve in first 48-72h)

Management:

- **IV fluids** are the only effective therapy for pancreatitis in the first 24h
 - Moderately aggressive fluid resuscitation (1.5mL/kg/h) w/ **Ringers Lactate** recommended as per results of WATERFALL trial 2022 with boluses PRN for hypovolemia (most will require 3-4L in first 24h)
 - Reassess response to IVF frequently (vitals, urine output, BUN, HCT) especially if cardiac and renal disease
- **Nutrition**
 - Early oral enteral nutrition within 24-48h with *low-fat solid diet* - If unable to take PO, NG tube preferred over NJ
- Prophylactic antibiotics are **not** recommended unless evidence of cholangitis or infected necrosis

Management of Pancreatic Necrosis

Suspect Infected Pancreatic Necrosis:

Typically, after day 7-10
Renewed pain, fever, SIRS

SKIP CT Aspiration!

Assume Infected Necrosis
Use Targeted Antimicrobial Therapy
Pancreatic Necrosis Penetrating Antibiotics

Clinically Stable

Continue Antibiotics and Observe...
Delay Minimally Invasive Surgical, Radiologic,
or Endoscopic Debridement 4-6 weeks.
In Select Patients who do well: No Debridement!

Clinically Unstable

Prompt Surgical
Debridement

- ACG 2024 skips previous step of obtaining CT guided FNA to distinguish between sterile and infected necrosis
 - Go straight to empiric antibiotics if suspecting infection!
- Preferred antibiotics = carbapenem or FQ/cephalosporin + metronidazole
 - Treat x 2-4 wks as necrosis collection becomes walled off and more amenable to endoscopic or surgical intervention

Approach to Acute Pancreatitis

- **Gallstone pancreatitis**
 - Early ERCP (<24 hours) recommended for cholangitis or jaundice
 - Rectal NSAID (i.e. indomethacin) + pancreatic duct stent to reduce risk of post ERCP pancreatitis
 - Medical therapy recommended over early ERCP in absence of cholangitis as most small stones will pass on their own without intervention
 - MRCP or EUS preferred to screen for persistent CBD stones over diagnostic ERCP
 - Ideally, cholecystectomy before discharge
- **Hypertriglyceridemia-induced pancreatitis**
 - Generally only if TG very high (> 11)
 - If sick → IV insulin, **strict NPO** +/- plasmapheresis
 - Long term → Fibrates, restrict dietary fat, treat secondary causes of hyperTG (diabetes, ETOH)

BONUS
Read on own

Classification and Complications of Acute Pancreatitis

The Atlanta Classification

Early (<4 weeks)

- Interstitial edematous pancreatitis (IEP)
 - Most patients with pancreatitis
 - Diffuse pancreatic edema
- Acute peripancreatic fluid collection
- Pancreatic necrosis
 - 5-10% patients
 - ↑ morbidity vs IEP
- Acute necrotic collection
- Infected pancreatic necrosis
 - Usually develops after 1 week

Late (≥ 4 weeks)

- Pseudocysts
 - Walled-off collection of pancreatic juices enclosed by fibrous tissue
 - 'Pseudo'cyst as no true epithelial lining
 - Can cause biliary or GI obstruction
 - Develop after ~4 weeks
 - Can become infected, requiring endoscopic/surgical drainage
- Pancreatic abscess
 - Circumscribed collection of pus w/o pancreatic necrosis
 - No longer recognized in the current Atlanta Classification
- Walled-off pancreatic necrosis
 - Consider endoscopic, surgical, or IR necrosectomy; observation in asymptomatic individuals

IgG4/Autoimmune Pancreatitis

There are 2 types of AIP

- **Type I** part of IgG4 RD which also has biliary, salivary, pulmonary, renal, thyroid, LN involvement
 - Mostly older men (> 60 yo), associated with elevated IgG4 level
- **Type II** isolated to pancreas and associated with IBD
 - M = F; > 40 yo

Both types of AIP can mimic pancreatic cancer in presentation and radiographic appearance

- But unlike cancer, responds to steroids!

Workup

- CT/MRCP + EUS for biopsy
 - “Sausage pancreas” on MRCP
 - Biopsy (EUS + FNA of pancreas or CT-guided Bx of retroperitoneal mass or other involved organ)
 - Histology = lymphoplasmacytic infiltrate + fibrosis
 - Biliary strictures (IgG4-associated cholangitis)
- Serum IgG4 > 2x ULN suggestive of Type I AIP (but can be negative!)

Treatment

- Prednisone 40mg daily x 4-6 weeks, then taper
 - If relapse → retreat steroids or AZA or Rituximab
- Look for radiologic improvement

Chronic Pancreatitis

- **Etiology: “TIGAR-O”**

- Toxic/metabolic: CHRONIC alcohol use, hyperTG, Smoking, hypercalcemia, Medications, Toxins
- Idiopathic
- Genetic – Cystic Fibrosis, and others PRSS1, CTFR...
- Autoimmune pancreatitis
- Recurrent Acute Pancreatitis (RAP)
- Obstructive: i.e. pancreatic divisum (2 distinct pancreatic ducts, not one), stricture, stone, tumor

- **Clinical Manifestations:**

- Severe abdominal pain
- Fat-soluble vitamin deficiency (ADEK)
- Exocrine pancreatic insufficiency → steatorrhea, maldigestion, weight loss
- Endocrine pancreatic insufficiency → Diabetes

- **Diagnosis:**

- 1st line – Cross-sectional imaging (CT or MRI) → EUS if uncertain
- Role of pancreatic exocrine function testing (eg 72h fecal fat) unclear in diagnosis of CP.

- **Work-up:**

- Ca²⁺, Triglycerides (≥ 5.6 mmol/L)
- Med review – i.e. azathioprine, cyclosporine, diuretics (furosemide, thiazide)
- IgG subclass – IgG4 disease
- Recurrent acute:
 - Consider MRCP – r/o pancreas divisum, stricture, tumor, stone
 - Consider genetics workup “if etiology unclear, especially in younger patients” – PRSS1, CFTR, CTFR, SPINK1

Chronic Pancreatitis - Treatment

- Lifestyle Modification:
 - Alcohol and smoking cessation for all
- Pancreatic enzymes
 - No evidence for relieving pain
 - Use for patients with exocrine pancreatic insufficiency
 - ↑ weight, ↑ fat-soluble vitamin absorption
- Pain management
 - Antioxidants (limited benefit) i.e. selenium, ascorbic acid, methionine
 - Opioids – if other pain control methods exhausted
 - Celiac plexus block - ↓ pain x 3-6 mo.
- Obstructive cause – trial of ERCP, EUS. If patient has chronic pain failing endoscopic therapy, “surgical intervention for long term pain relief”
 - CAVEAT: “Performing procedures on individuals using alcohol should be considered cautiously”
- Total pancreatectomy + islet cell autotransplant – highly select pts.

BONUS

Read on own

Choosing Wisely Canada

- 1) Don't maintain long-term PPI therapy for GI symptoms without an attempt to stop/reduce PPI once a year (Patients with Barrett's, severe esophagitis, or GI bleeding would be exempt)
- 2) Avoid using an upper GI series to investigate dyspepsia
- 3) Avoid performing an endoscopy for dyspepsia without alarm symptoms for patients under the age of 60 years
- 4) Avoid performing colonoscopy for constipation in those under the age of 50 without family history of colon cancer or alarm features
- 5) Do not routinely perform colonoscopy in IBS patients < 50 years of age without alarm features
- 6) Do not use steroids to maintain remission in IBD
- 7) Do not use opioids long-term for the management of pain in IBD
- 8) Don't use abdominal computed tomography (CT) scan to assess inflammatory bowel disease (IBD) in the acute setting unless there is suspicion of a complication (obstruction, perforation, abscess) or a non-IBD etiology for abdominal symptoms
- 9) Don't initiate or escalate long-term medical therapies for the treatment of inflammatory bowel disease (IBD) based only on symptoms.
- 10) Don't unnecessarily prolong the course of intravenous corticosteroids in patients with acute severe ulcerative colitis (UC) in the absence of clinical response.

Questions?

Good luck on your exam!



BONUS SLIDES

- PPI fast facts
- LGIB, SB bleeding
- IBD bonus slides
- Acute Cholangitis
- Irritable Bowel Syndrome
- Microscopic colitis
- Approach to Chronic Diarrhea
- Refeeding Syndrome
- ERCP/EUS
- AIH/ PSC/PBC
- Wilson's Disease
- Paracentesis + cirrhosis complications
 - Check out 2018 Online Video for this!
- Ogilvie's syndrome – NEW!
- SMA syndrome vs. MALS – NEW!
- BONUS MCQs

BONUS

Read on own

Lower GI Bleeding

- Rule out brisk UGIB or other source of bleeding
- DDx of rectal bleeding
 - **Painless**
 - Diverticular bleed (most common), hemorrhoids, angiodysplasia, polyp/post-polypectomy, cancer, ulcers (NSAIDs, stercoral), radiation proctitis, Meckel's diverticulum
 - **Painful**
 - Ischemic colitis (sudden, cramping abdo pain followed by bloody bowel movement)
 - Work-up cause: Cardio-embolic? (AFIB? Endocarditis? LV Thrombus) ATHERO-embolic (?AAA, dissection...), Shock/low flow, Segmental – think vasculitis / PAN
 - Risk factors for poor outcome: Male, sBP < 90, HR > 100, abdominal pain without rectal bleeding, BUN > 20 mg/dL, Hb < 120, LDH >, Na < 136, WBC > 15
 - Mgmt: Most resolve with supportive care (IV fluids, analgesia), empiric antibiotics if moderate-severe (strong recommendation, low quality evidence), Gen Sx if pancolitis or isolated right-sided
 - Hemorrhoid (if thrombosed)
 - Anal fissure
 - **Other** (may be associated with cramping)
 - Inflammatory (IBD), Infectious

ACG 2015 colonic ischemia guidelines: https://journals-lww-com.myaccess.library.utoronto.ca/ajg/fulltext/2015/01000/acg_clinical_guideline_epidemiology_risk.8.aspx

BONUS

Read on own

Lower GI Bleeding

- ABC-MOIF: (Monitor Oxygen IV, Foley) Resuscitation and management identical to UGIB
- Consider EGD for suspected rapid UGIB (usually BRBPR with hemodynamic instability, elevated urea)
 - NG tube can rule in UGIB (coffee grounds/blood per NG) but cannot rule it out
- Consider preparation for colonoscopy within 24 hours with full bowel prep
 - If recurrent bleeding consider repeating colonoscopy
- Early referral to surgery or IR for embolization (CT angiography) for unstable patients or those unable to tolerate colonic preparation
- Stop NSAIDs or primary prevention ASA

ACG Guideline: Management of Patients with Acute Lower Gastrointestinal Bleeding
(*Am J Gastroenterol* 2016; 111:459–474)

BONUS

Read on own

Small Bowel Bleeding

- Suspect in patients with ongoing overt GI bleeding with normal EGD and colonoscopy

Diagnostic algorithm if ongoing bleed despite negative EGD/ colonoscopy (in this order):

1. Second-look EGD +/- push enteroscopy (to look up to proximal jejunum) +/-colonoscopy
 2. Video capsule endoscopy (VCE)
 3. CT enterography (CTE)
 - CTE before VCE if concern for tumors, IBD, hx of radiation or chronic NSAID use which would be risk factors for strictures or other obstructive process where capsule could get stuck
- CTA if brisk, active bleeding; tagged RBC scan if slow bleed (>0.1 mL/minute)
 - Conventional angiography in IR suite for massive, unstable bleeding
 - Intraoperative enteroscopy (invasive; rarely done)

- **Treatment options:**

- Double-balloon enteroscopy, surgery, IR embolization
- Supportive management with transfusions and iron replacement

BONUS

Read on own

CD vs. UC

Disease features	Crohn's disease	Ulcerative colitis
Pathology	Transmural Non-caseating granulomas (characteristic but only in < 30% of pts) Lymphoid aggregates	Mucosal/submucosal Cryptitis/ architectural distortion, crypt abscesses, atrophy Shallow, broad-based ulcers
Distribution	Skip lesions, "gum to bum," UGI involvement possible (rare) Most common: ileocecal region	Contiguous colonic inflammation starts in rectum
Complications	Obstruction (strictures), fistula, abscess, perianal dz	Toxic megacolon
Cancer risk	Colorectal CA (colitis only)	Colorectal CA If PSC → risk of hepatobiliary cancers incl HCC, cholangiocarcinoma, GB cancer
Serology *routine use of serology is NOT recommended because poor SN/SP	ASCA (anti saccharomyces cerevisiae Antibodies), Anti-OmpC, Anti-CBir1	P-ANCA

BONUS
Read on
own

Drugs	Indication	Side Effects	Caution / CI
5-ASA	Little role in Crohn's UC: induction, maintenance -Suppository if up to 18cm disease only -Enema if beyond this but distal to splenic flexure	Headache, diarrhea, nausea, pancreatitis interstitial nephritis	Allergy
Steroids	Topical steroids (ie: budesonide) can be used longer-term	Less steroid S/E (high first pass effect)	Non-TI or R Colon disease
	Systemic steroids only in the short term – "steroid bridge"	Usual steroid S/E	
Azathioprine	Crohn's or UC maintenance	Hepatotoxicity, leukopenia, lymphoma	TPMT deficiency
Methotrexate	Crohn's induction/maintenance	Cytopenias, hepatotoxicity, ILD (See rheum slides for more)	Women of childbearing age
Anti-TNFs + other biologics (anti IL 12/23)	Tx of fistulizing disease, perianal disease, or moderate to severe disease	Infections, lymphoma, other malignancy	TB reactivation, HBV reactivation
Tofacitinib / Upadacitinib	Moderate-severe UC with loss of response/lack of response/intolerance of conventional therapy or anti-TNF	Infections, lymphoma, other malignancy, leukopenia, anemia, MACE, thrombosis (arterial/ venous)	CAD, thrombosis HZV, TB reactivation GI perforation
Antibiotics (Cipro/Flagyl)	Some evidence for use, but for RC purposes limit to Tx of abscess, active infections, or fistulas	Neuropathy (Flagyl) Achilles tendonitis (Cipro)	Allergy
S1P Receptor Modulators	Moderate-to-severe UC induction and maintenance	Bradycardias, infections, lymphopenia, elevated liver enzymes, macular edema (get baseline slit lamp in those w/ risk factors e.g. DM retinal disease)	C/I in patients with MI, unstable angina, CVA, NYHA 3/4 CHF in last 6 months; pregnancy; breastfeeding; Mobitz Type 2 or 3 rd degree heart block

IBD THERAPIES

BONUS

Read on own

Acute Cholangitis

Bacterial infection of the **biliary tree** due to **partial or complete obstruction**

Causes of Obstruction

- **Stones:** Choledocholithiasis, Mirizzi syndrome, sludge
- **Strictures:** Malignant (e.g. cholangiocarcinoma), benign
- **Tumors:** Pancreatic head, Klatskin tumor, metastases
- **Less common: Stent blockage, PSC, HIV cholangiopathy,**
- **Parasites:** *Ascaris*, tapeworms, **Clot**

Clinical Presentation

- **Charcot's Triad:** Fever, RUQ pain, jaundice
- **Reynolds' Pentad:** + Hypotension, confusion (severe disease)

Lab Findings

- WBC ↑ (>10) or ↓ (<4)
- ALP, GGT > 1.5× ULN, ALT > 1.5× ULN, Bilirubin > 34 μmol/L

Imaging

- **Biliary dilatation**, ± stones
- Modalities : US, CT, MRCP, Endoscopic US (most sensitive)

Investigations:

- CBC, lytes, creatinine, INR/PTT
- AST, ALT, ALP, GGT, total + direct bilirubin
- Blood cultures

Empiric Antibiotics:

- Cover common biliary pathogens – CTX/flagyl, piptazo, cipro/flagyl, carbapenem

'Source control'

- ERCP + stone removal +/- sphincterotomy
 - *↑ bleeding risk! Ensure INR < 1.5, PLT >50, anticoagulants held) +/- stent placement
- IR-guided percutaneous transhepatic biliary drainage – if failed ERCP or ++ comorbidities, bleeding risk etc.
- Surgical biliary drainage – rarely required

**↑ bilirubin + ALP are NOT common in acute cholecystitis.
Have a high index of suspicion for cholangitis!**

Ahmed M. Acute Cholangitis – An Update. [World J Gastrointest Pathophysiol](#). 2018 Feb 15; 9(1): 1–7.

BONUS
Read on own

Irritable Bowel Syndrome

- **Diagnosis:**

- Positive diagnostic strategy for IBS is favoured over IBS as a dx of exclusion (ACG 2021)
- ALL patients should be tested to exclude celiac disease (TTG-IgA + IgA)
- Recommendation is AGAINST routine colonoscopy unless patient is >50y or has alarm features

- **Treating global symptoms of IBS:**

- Low FODMAP Diet (fermentable oligosaccharides, disaccharides, monosaccharides and polyols)
- Soluble fibre e.g. psyllium (NOT insoluble fibre)
- Peppermint oil
- Cognitive behavioural therapy
- Gut-directed psychotherapy
- Low dose tricyclic antidepressants → be careful in patients with IBS-C as it may worsen constipation
- SSRIs (only recommended by CAG)
- Antispasmodics (ACG recommends *against*; CAG makes conditional recommendation for this based on very low quality of evidence)

- **Pharmacotherapy for IBS-C:**

- Linaclotide (Constella), Plecanatide, Tenapanor (Ibsrella), Lubiprostone (in Canada, indication is for chronic idiopathic constipation)

- **Pharmacotherapy for IBS-D:**

- Rifaximin, Eluxadoline (Viberzi) - C/I in those with prior cholecystectomy or hx of heavy EtOH use

Rome IV Criteria

Recurrent ***abdo pain*** at least once a week in the past 3 months w/ symptom onset 6 months before dx AND 2 of:

1. Related to defecation (improvement OR worsening of pain with BM)
2. Associated with change in stool frequency
3. Associated with change in stool form

CAG 2019 recommends against use of PEG for IBS-C and loperamide or cholestyramine for IBS-D while these therapies are often used in practice and also recommended by AGA 2022. Go with CAG guidelines for exam!

CAG 2019 Management of Irritable Bowel Syndrome (IBS). <https://doi.org/10.1093/jcag/gwy071>

ACG 2021 Management of Irritable Bowel Syndrome. <https://doi:10.14309/ajg.0000000000001036>.

AGA 2022 Clinical Practice Guideline on the Pharmacological Management of Irritable Bowel Syndrome With Diarrhea. <https://doi:10.1053/j.gastro.2022.04.017>.

AGA 2022 Clinical Practice Guideline on the Pharmacological Management of Irritable Bowel Syndrome With Constipation. <https://doi:10.1053/j.gastro.2022.04.016>.

BONUS

Read on own

Microscopic colitis

Two types: lymphocytic or collagenous colitis

- Epidemiology, risk factors
 - Older women (mean age 65)
 - Comorbid autoimmune diseases – thyroid, celiac, RA
 - Associated medications: NSAIDs, PPIs (**specifically lansoprazole**), SSRIs, pembrolizumab, others
- Symptoms
 - Relapsing/remitting watery, non-bloody diarrhea; may be severe
 - Weight loss, abdominal pain are common
- Diagnosis
 - Normal-appearing colonoscopy, biopsies confirm diagnosis
- Treatment
 - Stop NSAIDs/other offending meds, smoking cessation, Imodium prn
 - 1st line: Budesonide, consider prednisolone if refractory to budesonide
 - 2nd line: 5-ASA PO, bismuth salicylate

BONUS

Read on own

Approach to Chronic Diarrhea

- **Chronic diarrhea:** Increased stool frequency (>3 BM/day) or decrease in stool consistency for > 4 weeks
- **Etiology**
 - **Infection:** C difficile, CMV (suspect if immunocompromised!), yersinia, giardia, cryptosporidium, strongyloides, Whipples disease, TB, etc.
 - **Inflammatory:** IBD, microscopic colitis, radiation colitis, immune checkpoint inhibitor colitis
 - **Neoplastic:** Colon cancer, intestinal lymphoma, neuroendocrine tumours (e.g. VIPoma, gastrinoma), paraneoplastic diarrhea
 - **Endocrine:** hyperthyroidism, Addison's, autonomic neuropathy in setting of diabetes
 - **Malabsorption:** Lactose intolerance, celiac disease, tropical sprue SIBO, pancreatic insufficiency, post-operative (cholecystectomy, Roux-en-Y gastric bypass, small bowel resection – especially extensive ileal resections, pancreatectomy)
 - **Medication induced:** Antibiotics, metformin, colchicine, NSAIDs, PPIs, chemotherapy, Mg supplement, & more
 - **Other:** factitious diarrhea, fecal incontinence, IBS-D
- **Investigations**
 - CBC +/- ferritin → Fe deficiency anemia = alarm feature warranting colonoscopy
 - Electrolytes, urea, Cr, extended electrolytes → assess for complications/signs of volume depletion
 - TTG-IgA, IgA, TSH → r/o secondary metabolic causes (hyperthyroidism, celiac disease)
 - Stool studies
 - Stool cultures, c difficile, ova & parasites (especially giardia)
 - Fecal calprotectin → sens = 93%, spec = 96% for differentiating between inflammatory & non-inflammatory causes of diarrhea
 - Stool pH, stool Na+, stool K+, consider fecal elastase if suspicion of pancreatic insufficiency
- **Management**
 - Treat underlying cause
 - Supportive management: loperamide PRN, cholestyramine for bile acid diarrhea (CAG 2019)

Stool osmotic gap

$$= 290 - [\text{Stool Na} + \text{Stool K}] \times 2$$

> 100 suggests osmotic diarrhea while
< 50 suggests secretory diarrhea

BONUS

Read on own

Refeeding Syndrome

- **Definition:** Decrease in ≥ 1 of K, PO₄, and/or Mg or manifestations of thiamine deficiency within hours to days of rapid reintroduction of parenteral or enteral nutrition in an individual with inadequate nutritional intake
- **Who is at risk:** Chronic alcohol misuse, eating disorders, food insecurity/homelessness, refugees, cancer, post-operative patients, malabsorptive states [extensive list], dysphagia, dementia, low K/Phos/Mg at baseline
- **Clinical features:**
 - **Hypophosphatemia:** paresthesias, delirium, coma, seizures, tetany, dyspnea/respiratory failure, cytopenias
 - **Hypokalemia:** weakness, paralysis, respiratory failure, arrhythmias, nausea, vomiting, ileus/constipation
 - **Hypomagnesemia:** weakness, tremor, altered mental status, tetany, arrhythmias, seizures, nausea, vomiting, constipation
 - **Thiamine deficiency:** nystagmus, neuropathy, dementia, Wernicke-Kersakoff, beriberi
 - **Sodium retention:** edema
- **Treatment & Prevention:**
 - Monitor serum K, Mg, PO₄ while initiating and advancing nutrition, check baseline ECG, consider telemetry
 - Monitor vitals, cardiorespiratory status, weight, intake and output
 - Start feeds at 50% of energy requirements and advance slowly [consult dietitian]
 - Supplementation: Multivitamin, B complex, and Thiamine daily
 - Consider other labs for nutrients in those with risk (e.g. iron, b12, vit D etc.)

BONUS

Read on own

Autoimmune Hepatitis (AIH)

	Type 1 AIH	Type 2 AIH
Antibodies	ANA > 1:80, ASMA > 1:80 ANA, AAA, Anti-soluble liver/ liver pancreas antigen	Anti-LKM1 > 1:80 Anti-liver cytosol
Age of presentation	Any age	Children and young adults
Associated disorders	Autoimmune thyroid disease, Grave's, UC, celiac, T1DM	Autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED)

Clinical presentation:

- F >> M (5:1)
- May be asymptomatic, or present with acute hepatitis, acute liver failure, r chronic hepatitis or cirrhosis
- Drug-induced AIH (e.g. infliximab, adalimumab, nitrofurantoin, minocycline, atorvastatin, others)

Diagnosis:

- Constellation of clinical, lab (+ serology and elevated IgG), and histology (interface hepatitis)
- Exclude other causes of chronic hepatitis (NASH, viral, hereditary etc.)
- Can use the simplified AIH score to aid diagnosis

Treatment – Depends on presentation severity and presence of cirrhosis:

- Typically steroids (prednisone or budesonide) +/- azathioprine (not used in severe acute AIH)
*See excellent chart in guidelines
- Screen all patients for celiac disease and thyroid disease

BONUS

Read on own

Primary Sclerosing Cholangitis (PSC)

- Chronic, cholestatic, immune-mediated disease of intra & extrahepatic bile ducts (radiologic diagnosis)
- **Clinical presentation:**
 - M >> F, 2/3rd of patients have IBD (most have ulcerative colitis phenotype).
 - Up to 50% asymptomatic at presentation. Symptoms include abdominal pain, pruritus, fatigue.
- **Diagnosis:**
 - (1) Elevated ALP, (2) multifocal biliary strictures “beads on a string” (usually on MRCP), (3) exclude secondary sclerosing cholangitis, (4) liver Bx if PSC-AIH overlap of small duct PSD suspected.
- **Differential diagnosis:**
 - IgG4 related disease (exclude since this entity is steroid-responsive)
 - PSC-autoimmune overlap syndrome (respond partially to immunosuppressants)
- **Treatment:**
 - ERCP as needed for symptomatic strictures
 - MRCP +/- CA 19-9 q1 year to screen for GB cancer and cholangiocarcinoma
 - Colonoscopy with surveillance biopsies at diagnosis and q1 year (increased risk of colorectal cancer)
 - If cirrhotic, screen for HCC & varices
 - Ursodeoxycholic acid has **NO** evidence but frequently used; liver transplant for liver failure

BONUS

Read on own

Primary Biliary Cholangitis (PBC)

Disease of intrahepatic biliary tree

- **Clinical presentation**
 - Usually middle-aged women (>95% of cases)
 - Present with jaundice, pruritus, fatigue, elevated ALP (usually get imaging – either U/S or MRCP to rule out obstruction – will be normal in PBC)
 - Associated with Sjogren's, celiac disease, and autoimmune thyroid disease
- **Diagnosis requires 2 of 3:**
 1. Persistent ALP elevation > 6 months
 2. Positive AMA >1:40 (95% sensitive, 98% specific) or specific ANAs (e.g. gp 210/sp 100)
 3. Liver biopsy (only needed when diagnosis unclear)
- **Management/Treatment**
 - Bone density testing, lipid profile derangements, monitor for cirrhosis
 - Ursodeoxycholic acid 15mg/kg has benefit and remains 1st line
 - Second line: Obeticholic acid (avoid in decomp cirrhosis), fibrates (used off label)
 - New agents: Seladelpar (PPAR- δ agonist), Elafibrinor (PPAR α/δ agonist) – second line if non responsive or intolerant to UDCA
 - Liver transplant for decomp cirrhosis or treatment refractory disease

BONUS

Read on own

Wilson's Disease

Etiology:

Autosomal recessive mutation in **ATP7B** gene → ↓ biliary copper excretion → copper accumulation (liver, brain, other organs)

Clinical Features

- **Hepatic:** Elevated LFTs, steatosis, cirrhosis, acute liver failure
- **Neurologic:** Dysarthria, tremor (postural > rest), dystonia, parkinsonism
- **Psychiatric:** **typically onset in 2nd, 3rd decade of life may precede neurologic signs**
Mood/behavioral changes, psychosis
- **Other:**
 - **KF rings**, sunflower cataracts
 - **Hemolytic anemia (Coombs-negative)**
 - Renal (Fanconi), cardiac, bone, infertility

Diagnosis

- ↓ **Ceruloplasmin** (may be normal in inflammation/acute phase)
- ↑ **24h urinary copper**
- **Genetic testing (ATP7B)**
- **Liver biopsy** if unclear
- **Slit lamp** (KF rings), **MRI brain** (neurologic symptoms)

Treatment

- **Chelators:** D-penicillamine or trientine (*lifelong*)
- **Zinc:** 1st line for asymptomatic cases
- **Monitor:** LFTs, ceruloplasmin, urinary copper
- **Transplant:** If decompensated cirrhosis
- Screen 1st-degree relatives
- HCC surveillance if cirrhotic



BONUS
Read on own

Paracentesis (Oral Scenario)

- **Diagnostic**
 - Cell count, culture and sensitivity, biochemistry including albumin (to calculate SAAG), cytology, AFB, etc.
 - Risks: bleeding, infection, perforation, organ injury
- **Therapeutic**
 - Purpose is to reduce ascites for symptomatic patients
 - Can take off unlimited amount of fluid as one time (suggest replace with 100cc of 25% albumin for each 4L of fluid removed)
 - Avoid therapeutic paracentesis for patients with SBP, UGIB, active infection, or acute kidney injury (may worsen renal function)
 - Risks are same as diagnostic paracentesis **AND** risk of hypotension, AKI, encephalopathy

BONUS
Read on own

Hepatic Encephalopathy (HE)

- **Etiology**
 - Accumulation of glutamine from increased ammonia levels
 - Patients can present with day-night sleep reversal, drowsiness, agitation, confusion
- **Diagnosis**
 - Clinical diagnosis! (no role for ammonia levels)
 - Look for precipitating causes (SBP, infection, GI bleeding, alcohol, drugs, HCC, stroke, constipation, dehydration, portal or hepatic vein thrombus, electrolyte abnormalities, TIPS)
- **Treatment**
 - Treat underlying cause (stop benzos, opioids, anti-chol)
 - **Lactulose +/- Rifaximin (typically added if patient had episode of HE while on lactulose)**
 - Adjunctive therapies: Zinc, branch-chain amino acid supplementation, late evening protein snacks (to prevent catabolism which indirectly worsens HE)
 - No need to restrict dietary protein intake
 - Recurrent intractable HE, together with liver failure, is an indication for liver transplant

BONUS

Read on own

Hepatorenal Syndrome

- **HRS:** Functional renal injury in pts with portal HTN 2^o cirrhosis/ acute cause (alc hep, fulminant liver failure)
- **Pathophysiology:** Portal HTN → splanchnic arterial vasodilation → AKI
- **HRS classification** (based on the rapidity of renal impairment):

HRS-AKI (formerly type 1 HRS): More severe, abrupt (<48h) decline in renal function in pt w/ no prior renal disease
Diagnostic Criteria:

1. Cirrhosis/ acute portal HTN with ascites
2. $\uparrow$ Cr $\geq$ 0.3 mg/dl ($\sim$ 26 μ mol/L) or $\uparrow$ Cr $\geq$ 50% (from b/l within past 3 mo) or UO < 0.5 ml/kg/h for $\geq$ 6h.
3. Other causes ruled out (ATN, shock, drugs, obstructive)
4. No improvement with diuretic withdrawal + 1g/kg IV albumin x 2 days
5. Suggestion of renal vasoconstriction with a FENa < 0.2% (with levels < 0.1% being highly predictive)

HRS-NAKI [non-AKI] (formerly type 2 HRS): less severe, gradual decline in kidney function assoc. w/ refractory ascites

- **HRS-AKD** (HRS-acute kidney disease) if eGFR < 60 mL/min/1.73m² for < 3 months
- **HRS-CKD** (HRS-chronic kidney disease) if eGFR < 60 mL/min/1.73m² for > 3 months

- **Treatment**

- Definitive = liver transplant
- ICU = Norepinephrine and IV albumin
- Ward = Terlipressin + Albumin preferred over Octreotide + Midodrine + Albumin
 - Terlipressin approved by FDA but not widely available in Canada (**CONFIRM** trial 2021)

BONUS
Read on own

Hepatopulmonary Syndrome

- **Etiology**
 - Patients with cirrhosis or portal hypertension can get hypoxemia from intrapulmonary vasodilation
 - Classic presentation: dyspnea, platypnea, hypoxemia
- **Diagnosis:**
 - Suspect HPS if ABG reveals $\text{PaO}_2 < 80$, A-a gradient ≥ 15
 - Echo with agitated saline (i.e. bubble study) demonstrates evidence of intrapulmonary shunting
 - Need to exclude other causes
- **Treatment**
 - Supplemental oxygen, liver transplant reverses HPS

BONUS

Read on own

Ogilvie's Syndrome-NEW

- aka Acute Colonic Pseudo-obstruction (ACPO)= massive colonic dilation without mechanical blockage
- **Etiology:** drugs (narcotics, anticholinergics, CCV, TCA, antipsychotics), electrolyte disturbances, neurologic disease (PD, spinal cord injury), post op, trauma, sepsis, hypothyroidism
- **Complications:** ischemia, peritonitis, perforation risk (esp. if cecal diameter > 12 cm)
- **Management (as per ASGE 2020):**
 - If cecal diameter < 12 cm: (first 72 h) treat underlying cause, correct lytes, hold offending meds; NPO, NG tube to decompress, ambulate or roll patient periodically in bed
 - **If cecal diameter > 12 cm OR failure to improve within 72h :** Neostigmine 2- 5 mg IV slowly over 5 minutes (monitor ECG, consider atropine at bedside) and reassess. If failed OR if contraindications to neostigmine* → consider endoscopic decompression.
 - **Refractory cases → direct to surgery or percutaneous cecostomy tube**

NEOSTIGMINE*

Mechanism of action:

- Acetylcholinesterase inhibitor

Adverse Effects:

- BRADYCARDIA!
- Bronchoconstriction
- Nausea/Vomiting
- Diarrhea
- Salivation/drooling
- Diaphoresis
- Tremors
- Seizures

Absolute Contraindications:

- Bowel obstruction
- Bowel perforation
- Urinary obstruction
- Hypersensitivity reaction

Relative Contraindications

- Recent MI
- Bradycardia
- Hypotension (sBP<90)
- Renal insufficiency
- Hx of bronchospasm / asthma / reactive airways
- Seizure Disorder
- Peptic ulcer disease

BONUS

Read on own

SMA Syndrome vs. MALS- NEW

	Median Arcuate Ligament Syndrome (MALS) aka Celiac Artery Compression Syndrome (CACS)	Superior Mesenteric Artery Syndrome (SMA Syndrome)
Pathophysiology	Compression of the celiac artery by the median arcuate ligament of the diaphragm.	Compression of the third part of the duodenum between the aorta and SMA.
Patient Profile	Middle-aged adults (common in females).	Young, underweight individuals (often post-weight loss*).
Symptoms	- Postprandial epigastric pain (20-30 min after meals). - Pain relieved by leaning forward. - Weight loss. - Systolic bruit on exam.	- Postprandial epigastric pain (immediate after meals). - Pain relieved by lying prone or knee-chest position. - Nausea and vomiting. - Weight loss.
Key Clinical Sign	Systolic abdominal bruit (non-lateralized).	Pain relieved in prone or knee-chest position.
Imaging Findings	- Duplex ultrasound: Elevated celiac artery velocity on expiration. - CTA/MRA: Celiac artery narrowing.	- CT abdomen: Narrowing of the aortomesenteric angle (<25°). - Dilation of proximal duodenum.
Best Diagnostic Test	- Duplex ultrasound (dynamic with respiration). - Confirm with CT/MR angiography.	- CT abdomen with contrast for aortomesenteric angle. - Upper GI series for duodenal compression.
Treatment	- Initial: Dietary modifications (small meals). - Definitive: Surgical release of the median arcuate ligament.	- Initial: Nutritional support and weight gain. - Definitive: Surgical intervention (e.g., duodenojejunostomy) in refractory cases.

BONUS

Read on own

Bonus Question 1

You are seeing a 24-year-old female in your clinic with known perianal fistulizing Crohn's disease who has been having worsening symptoms of perianal pain and drainage. Which of the following would not be part of your work-up / management?

- A) Refer to general surgery
- B) Obtain an MRI pelvis
- C) Start oral mesalamine and rectal 5-ASA suppositories
- D) Refer for TB skin test and TPMPT testing

BONUS

Read on own

Bonus Question 2

A 43yM with Down's Syndrome lives in a group home setting. He has hyperphagia and is known to eat large quantities of food from kitchen particularly when stressed; this has worsened with pandemic related restrictions on activities and visitors. Group home staff say it is difficult to police his foods. He presents with a 2-month history of progressive severe pruritic blistering rash on his knees, hands, buttocks, elbows (inset). Which of the following tests would you not initiate?

- a) TTG-IgA
- b) HLA-DQ2, -DQ8 testing
- c) G6PD Assay
- d) Skin biopsy
- e) Skin scraping for microscopy



BONUS

Read on own

Bonus Question 3

A 29F @ 6 weeks pregnant with a new diagnosis of ulcerative pancolitis is admitted to hospital with a flare. She was admitted with worsening bloody diarrhea up to 7 BMs/ day, and was found to a Hgb 101, WBC 13, and Plt 489. Stool studies were negative and she was started on IV methylprednisolone on day 2 of admission. On Day 4 of admission, her nurse calls you for fever with a Tmax 38.1C, HR 120, and BP 100/62. Bedside Ultrasound: No free air. Dilated loops of bowel with maximal diameter of 7cm at the transverse colon.

What is the best next step?

- a) Give her an induction dose of infliximab
- b) Consult General Surgery
- c) Schedule sigmoidoscopy with biopsies to rule out CMV colitis
- d) Start IV ciprofloxacin and flagyl
- e) Start Lovenox 40mg daily thromboprophylaxis

BONUS
Read on own

Bonus Question 4

67F with a recent STEMI was started on clopidogrel and ASA. She had been on ranitidine 150 mg PO for years after a remote UGIB related to an H. pylori positive ulcer. She remembers receiving antibiotics for her H. pylori. She has had no recent GI bleeding. Which of the following do you recommend for her therapy?

- A. Continue ranitidine
- B. Double ranitidine dose to 300 mg
- C. Change ranitidine to a pantoprazole
- D. Treat for H pylori with PBMT

BONUS

Read on own

Bonus Question 5

48M with a 15-year history of ileocolonic Crohn's disease is admitted with a flare. He is not responsive to high dose IV steroids and plan to start infliximab as he does not want surgery. He had his normal vaccinations in childhood and has not had any vaccinations since then. What would you do in regards to vaccinations?

- A. Provide an MMR booster Now
- B. Provide the TdAP booster when steroids tapered to <20mg/d pred
- C. Provide a varicella booster when steroids tapered to <20mg/d pred
- D. Provide the intranasal influenza vaccine Now

BONUS

Read on own

Bonus Question 6

A 53F presents with 3 months of epigastric pain and 2 weeks of intermittent loose stools and melena. EGD shows diffuse gastric antrum and duodenal bulb ulcers. What would be the next best test?

- A. Secretin stimulation test
- B. Gastrin level
- C. H. Pylori IgG
- D. Therapeutic endoscopy

BONUS

Read on own

Bonus Question 7

66F presents with a one day history of rectal bleeding and pre-syncope. She describes a history of acute-onset left lower quadrant pain, followed by 4-5 episodes of bloody diarrhea. Past medical history is significant for T2DM, HTN, GERD and cigarette smoking. On exam, BP 88/54, HR 105. She is afebrile. She has mild guarding without rebound tenderness in the left lower quadrant. Hb 100, WBC 13, PLT 160. Stool C&S and c.diff were negative.

A CT performed in the ED reveals segmental colitis involving the transverse and descending colon without evidence of perforation or diverticulitis.

Flexible sigmoidoscopy revealed well-demarcated edematous, friable colonic mucosa with linear ulcerations from the sigmoid colon to splenic flexure, with relative rectal sparing.

Based on the likely diagnosis and current guidelines, how should she best be managed?

- A. IV heparin bolus + infusion
- B. IV fluids, cipro + flagyl, general surgical consultation
- C. ASA 81mg PO daily
- D. Supportive care – IV fluids, bowel rest, pain control

BONUS

Read on own

Bonus Question 8

30M hospitalized for 4 days with his first episode of UC. Vital signs normal. Flexible sigmoidoscopy shows severe colitis to 40 cm from the anal verge. Stool C+S, c diff are negative. Hb 90, WBC 16, Plt 460, Alb 21, CRP 42. No improvement on IV steroids x 3 days, 6-10 BMs/ 24h, > 50% bloody. Abdomen tender but not peritonitic. AXR: No free air, max colon diameter 4 cm.

What is the best next step?

- A. Increase the steroid dose
- B. Start infliximab
- C. Subtotal colectomy
- D. Total proctocolectomy

BONUS

Read on own

Bonus Question 9

46F with T2DM A1C 6.4%, arthritis, obesity, presenting with fatigue. Her Family MD calls for advice about finding of ferritin 680, TSat 38%, ALT 38, creatinine and CBC normal. Liver enzymes are within normal limits. Nothing on physical exam to explain fatigue except for poor exercise tolerance due to obesity. What do you advise for next steps?

- A. Test for HFE gene mutation
- B. Liver ultrasound
- C. Phlebotomy
- D. None of the above

BONUS

Read on own

Bonus Question 10

58F with 3 month history of watery diarrhea, up to 6 bowel movements per day. 3 BMs per night. Otherwise systemically well with no weight loss. She has a past medical history of depression, on escitalopram. Stool cultures and c diff are negative. Basic laboratory investigations negative. Anti-TTG is negative with a normal IgA level. ANA 1:160. CT abdo/pelvis normal. EGD and colonoscopy show normal appearing mucosa. Biopsies from the right and left colon are pending.

Which of the following is most likely to explain this patient's symptoms?

- A. Vasoactive peptide secreting tumour (VIP-oma)
- B. Pancreatic insufficiency
- C. Crohn's Disease
- D. Microscopic colitis

BONUS

Read on own

Bonus Question 11

63F with alcohol-related cirrhosis, decompensated with ascites. She presents with dyspnea on exertion. BP 92/54, HR 76, SpO2 95% on 2L NP. There is no variation of oxygenation with a change in position. Chest is clear to auscultation. JVP is at 3 cm above the sternal angle. Normal S1 S2. Holosystolic murmur. Abdomen is distended with moderate ascites. CXR showed mild costophrenic angle blunting. TTE shows TR, mild right ventricular hypertrophy. RVSP 55 mmg Hg.

Which of the following is true?

- A. This patient should be referred for right heart catheterization.
- B. This patient would benefit from a TIPS.
- C. An ABG is required for the diagnosis of this condition.
- D. Liver transplant will always reverse this patient's condition.

Bonus Question 12

A 56yo male is brought to the ED by EMS with frank hematemesis. He has a history of an NSTEMI with previous PCI to the LAD 2 years ago. He also has T2DM, hypertension, dyslipidemia and is on ASA, metformin, empagliflozin, bisoprolol, candesartan, and rosuvastatin. He has been adherent to all of his medications. His vitals on presentation are: HR 124, BP 88/56, O2 95%, T 36.8, RR 16. He is lethargic and confused but does rouse to verbal commands. His initial bloodwork shows: Hgb 62, MCV 103, WBC 8.2, Plt 118.

Which of the following is not part of your initial management?

- a) Administer 1g tranexamic acid
- b) Cross match, group & screen, and transfuse packed red blood cells
- c) Start IV pantoprazole, octreotide, and ceftriaxone
- d) Consult ICU for consideration of airway protection and admission to monitored bed

Bonus Question 13

A 62-year-old female from is seen in the urgent medicine clinic for a three-month history of progressive dysphagia to liquids and solids. She has had 40 lbs of unintentional weight loss over the last 3 months. She does not endorse any GI bleeding or constitutional symptoms. Which of the following would not be part of your initial work-up?

- a) Serology for chagas disease
- b) Referral to GI for EGD
- c) CT Chest / Abdomen
- d) BW to assess CBC and nutritional markers

Bonus Question 14- NEW 2026

A 33M originally from Nigeria, is diagnosed with chronic hepatitis B infection (HBsAg+, HBV DNA 1500 IU/ml). He has no evidence of advanced fibrosis on Fibroscan (F0-F1). Normal LFTs and biochemical parameters. He has no family history of hepatocellular cancer (HCC). What is the most appropriate recommendation regarding HCC surveillance in this patient?

- A. Abdominal US + AFP q 6 months starting at age of 40
- B. Abdominal US + AFP q 12 months starting at age of 50
- C. Abdominal US + AFP q 6 months starting immediately
- D. No HCC surveillance required until cirrhosis develops

Bonus Question 15 – NEW 2026

A 72-year-old man with multiple comorbidities develops progressive abdominal distension after hip surgery. He is diagnosed with acute colonic pseudo-obstruction (Ogilvie's syndrome). He has been managed conservatively with IV fluids, NG decompression, correction of electrolytes, and NPO. After 48 hours, there has been no clinical improvement and his cecal diameter is 12 cm. He remains hemodynamically stable and has no peritoneal signs. Lactate is normal. What is the next best step in management?

- A. Endoscopic decompression
- B. Neostigmine
- C. Refer for surgical intervention
- D. Nothing further management required

Bonus Question 16 - NEW 2026

A 55-year-old man with cirrhosis is on furosemide and spironolactone. He presents with progressive ascites and weight gain. Lab studies: 24-hour urine sodium 85 mmol/day, SAAG 15, serum creatinine stable. Other labs are unremarkable. What is the cause behind his worsening ascites?

- A. Non-compliance with sodium restriction
- B. Hepatorenal syndrome (HRS)
- C. Diuretic resistance
- D. Overdiuresis

Bonus Question 17- NEW 2026

A 36-year-old male presents with fatigue, dark urine, and generalized itching. He has no significant medical history and denies alcohol use or travel. Laboratory results reveal ALT 220 U/L, AST 180 U/L, total bilirubin 95 $\mu\text{mol/L}$, alkaline phosphatase 280 U/L, and IgG 20 g/L. ANA and anti-smooth muscle antibodies are positive. Ultrasound shows a normal liver without biliary obstruction. What is the best next step in management?

- A. Start ursodeoxycholic acid for cholestasis
- B. Begin prednisone and azathioprine for autoimmune hepatitis
- C. Perform an urgent ERCP
- D. Obtain a liver biopsy to confirm the diagnosis

BONUS
Read on own

BONUS MCQ Answers

Question 1 – C) While 5-ASA is an appropriate 1st line Tx for UC, it is NOT recommended for mod-severe Crohn's disease, especially with fistulizing phenotype. Referral to general surgery (A) would facilitate an EUA which is a primary diagnostic modality alongside MRI pelvis (B). Infliximab and azathioprine are 1st line Tx for perianal fistulizing CD so it would be a good idea to send the patient for pre-emptive biologic screening and TPMT testing

Question 2 – B) This is a high probability scenario for Celiac disease (DH!). Living in a group home with itchy rash, also need to think about scabies which can clinically appear as pruritic papular rash in same distribution so scraping for microscopy not a bad idea (absence of burrows on exam also helpful). Skin biopsy will be diagnostic of DH. G6PD assay needed as this pt will need dapsone – adherence to strict gluten free diet will be a challenge.

Question 3 – B) Patient is developing toxic megacolon and should receive an urgent general surgery referral. Don't give a blood thinner until you know she's not going for surgery - reasonable to start once Gen Surg has seen if not for OR tonight!). Don't give fluoroquinolones in pregnancy when other reasonable alternatives available (eg could give her Pip/Tazo, Ceft/Flagyl...) In versions of this question where you have to choose between antibiotics and surgery as next step, we feel the point of question is know when to call surgeon, you would also give antibiotics in real life!

Question 4– C) Change to pantoprazole. PPIs prevent GI bleeding in pts on DAPT (COGENT trial). No evidence that H2 blockers prevent GI bleeding in pts on DAPT. Some studies show interaction of clopidogrel with esomeprazole/ omeprazole, but not pantoprazole.

Question 5– B) Do not give live vaccines (MMR, intranasal flu, Shingrix) on TNFi without holding per ACR guideline and as he's in flare don't delay treatment to give live vaccines. For non live attenuated vaccines ACR (<https://www.rheumatology.org/Portals/0/Files/Vaccinations-Guidance-Summary.pdf>) suggests deferring these until on <20mg/pred per day (except NON LIVE (e.g. IM) influenza, give that anytime) Review 2021 CAG Guidelines for Immunization in IBD <https://doi.org/10.1093/jcag/gwab015> "In patients with IBD who require urgent immunosuppressive therapy, treatment should not be delayed in order to provide vaccinations."

BONUS
Read on own

Bonus MCQ Answers

Question 6 – B) Gastrin level. Suspect Zollinger-Ellison syndrome with numerous ulceration, esp in duodenal bulb

Question 7 – B) Severe ischemic colitis. Guidelines recommend antibiotics in addition to IV fluids.

Question 8 – B) Maximize medical therapy first. Provide infliximab (or cyclosporine (not usually used in Canadian institutions)) for steroid refractory acute severe UC. If not responding to infliximab within 5-7 days, consider surgical involvement.

Question 9 – D) Answer is none of the above. Variants of this question exist in Q bank. This patient can be screened for MASLD with clinically significant fibrosis- can be done by calculating a FIB-4 score and if needed refer for fibroscan. No indication to test for HFE gene as TSAT < 45%. Liver US can comment on steatosis, but not recommended as screening tool for MASLD as it does not assess fibrosis extent (unless it is performed with elastography). Additionally, in this stem, in the absence of abnormal LFTs/liver enzymes, no indication to arrange for liver US either way.

Question 10 – D) Microscopic colitis. Typically in females 55-70 yo. Watery diarrhea, can be significant. Sometimes abdominal pain and wt loss. Colonoscopy reveals normal mucosa. Diagnosis on biopsies of R and L colon. Risk factors: smoking, medications (PPIs esp lansoprazole, ranitidine, SSRIs, statins, NSAIDs, ASA). Autoimmune conditions may exist in up to 1/3rd of patients. Mgmt involves stopping meds that could be related. First line treatment is PO budesonide. Second line can consider 5-ASA, though off label.

Question 11 – A) Suspect portopulmonary hypertension (PPH) in this patient, defined as pulmonary arterial hypertension associated with portal hypertension (which we know this patient has because of ascites), with or without liver disease. Diagnosed on RHC (see resp slides). PPH rarely reverses with liver transplantation (whereas hepatopulmonary syndrome commonly reverses with transplant). MPAP > 50 mm Hg is considered a contraindication to liver transplantation because the mortality is 100%. TIPS would increase right ventricular and PA pressure and worsen this patient's condition. This is also why TIPS is contraindicated with right heart failure.

BONUS
Read on own

BONUS MCQ Answers

Question 12– A) TXA is NOT recommended for management of UGIB as per HALT-IT Trial given no benefit and possible harm

Question 13– A) While chagas can mimic achalasia which would be on the differential for dysphagia to solids & liquids, this is probably a jump in someone with undifferentiated dysphagia who has had no other imaging or work-up. The #1 priority in someone with rapidly progressive dysphagia and weight loss is to r/o malignancy through imaging and endoscopy

Question 14 – C) New CASL 2025 guidelines recommend surveillance starting at age ≥ 30 for those of African ancestry, regardless of cirrhosis. This patient is 33, so screening should begin now. HCC surveillance is always q 6 monthly US + AFP.

Question 15 – B) ASGE 2020 guidelines would recommend medical therapy with neostigmine before endoscopic decompression, provided no contraindications no neostigmine.

Question 16 – A) 24 hour urine sodium > 78 mmol/day and persistent worsening ascites shows that he is able to excrete sodium, and most likely reflects continued high dietary sodium intake.

Question 17– D) This patient's labs ($\uparrow$ ALT/AST, $\uparrow$ IgG, ANA+, SMA+) are highly suggestive of autoimmune hepatitis, but biopsy is required to confirm before starting treatment. Option A (ursodeoxycholic acid) is for cholestatic disease like PBC, not autoimmune hepatitis. Option B (prednisone + azathioprine) is indeed the mainstay of treatment, but should only be initiated *after biopsy confirmation*. Option C (ERCP) is unnecessary since ultrasound shows no biliary obstruction. Thus, the best next step is liver biopsy (histologic confirmation is needed for AIH)